



# IEEE Standard for Ethernet

## Amendment 6: Media Access Control (MAC) Service Interface and Management Parameters to Support Improved Precision Time Protocol (PTP) Timestamping Accuracy

IEEE Computer Society

Developed by the  
LAN/MAN Standards Committee

### **IEEE Std 802.3cx™-2023**

(Amendment to IEEE Std 802.3™-2022  
as amended by IEEE Std 802.3dd™-2022,  
IEEE Std 802.3cs™-2022, IEEE Std 802.3db™-2022,  
IEEE Std 802.3ck™-2022, and IEEE Std 802.3de™-2022)

**IEEE Std 802.3cx™-2023**  
(Amendment to IEEE Std 802.3™-2022  
as amended by IEEE Std 802.3dd™-2022,  
IEEE Std 802.3cs™-2022,  
IEEE Std 802.3db™-2022,  
IEEE Std 802.3ck™-2022,  
and IEEE Std 802.3de™-2022)

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Developed by the  
**LAN/MAN Standards Committee**  
of the  
**IEEE Computer Society**

Approved 30 March 2023  
**IEEE SA Standards Board**

**Abstract:** This amendment to IEEE Std 802.3-2022 modifies Clause 30, Clause 45, and Clause 90, and adds Annex 90A to enhance support for time synchronization protocols by providing options for sub-nanosecond reporting of the transmit and receive path data delays, selection of the data delay measurement point, and dynamic reporting of path data delay variation.

**Keywords:** amendment, IEEE 802.3™, IEEE 802.3cx™, timestamp

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# Introduction

This introduction is not part of IEEE Std 802.3cx-2023, IEEE Standard for Ethernet—Amendment 6: Media Access Control (MAC) Service Interface and Management Parameters to Support Improved Precision Time Protocol (PTP) Timestamping Accuracy.

IEEE Std 802.3™ was first published in 1985. Since the initial publication, many projects have added functionality or provided maintenance updates to the specifications and text included in the standard. Each IEEE 802.3 project/amendment is identified with a suffix (e.g., IEEE Std 802.3ba™-2010).

The half duplex Media Access Control (MAC) protocol specified in IEEE Std 802.3-1985 is Carrier Sense Multiple Access with Collision Detection (CSMA/CD). This MAC protocol was key to the experimental Ethernet developed at Xerox Palo Alto Research Center, which had a 2.94 Mb/s data rate. Ethernet at 10 Mb/s was jointly released as a public specification by Digital Equipment Corporation (DEC), Intel and Xerox in 1980. “Local Area Networks: Carrier sense multiple access with collision detection (CSMA/CD) access method and physical layer specifications” was approved as an IEEE standard by the IEEE Standards Board in 1983 and subsequently published in 1985 as IEEE Std 802.3-1985. Since 1985, new media options, new speeds of operation, and new capabilities have been added to IEEE Std 802.3. A full duplex MAC protocol and the ability to use an EtherType to specify the MAC client protocol were added in 1997. The title was changed to Standard for Ethernet with the 2012 Revision.

Some of the major additions to IEEE Std 802.3 are identified in the marketplace with their project number. This is most common for projects adding higher speeds of operation or new protocols. For example, IEEE Std 802.3u™ added 100 Mb/s operation (also called Fast Ethernet), IEEE Std 802.3z™ added 1000 Mb/s operation (also called Gigabit Ethernet), IEEE Std 802.3ae™ added 10 Gb/s operation (also called 10 Gigabit Ethernet), IEEE Std 802.3ah™ specified access network Ethernet (also called Ethernet in the First Mile) and IEEE Std 802.3ba added 40 Gb/s operation (also called 40 Gigabit Ethernet) and 100 Gb/s operation (also called 100 Gigabit Ethernet). These major additions are all now included in and are superseded by IEEE Std 802.3-2022 and are not maintained as separate documents.

At the date of IEEE Std 802.3cx-2023 publication, IEEE Std 802.3 was composed of the following documents:

## IEEE Std 802.3-2022

Section One—Includes Clause 1 through Clause 20 and Annex A through Annex K and Annex 4A. Section One includes the specifications for 10 Mb/s operation and the MAC, frame formats and service interfaces used for all speeds of operation.

Section Two—Includes Clause 21 through Clause 33 and Annex 22A through Annex 33A. Section Two includes management attributes for multiple protocols and speed of operation as well as specifications for providing power over twisted pair cabling for multiple operational speeds. It also includes general information on 100 Mb/s operation as well as most of the 100 Mb/s Physical Layer specifications.

Section Three—Includes Clause 34 through Clause 43 and Annex 36A through Annex 43C. Section Three includes general information on 1000 Mb/s operation as well as most of the 1000 Mb/s Physical Layer specifications.

Section Four—Includes Clause 44 through Clause 55 and Annex 44A through Annex 55B. Section Four includes general information on 10 Gb/s operation as well as most of the 10 Gb/s Physical Layer specifications.

Section Five—Includes Clause 56 through Clause 77 and Annex 57A through Annex 76A. Clause 56 through Clause 67 and Clause 75 through Clause 77, as well as associated annexes, specify subscriber access and other Physical Layers and sublayers for operation from 512 kb/s to 10 Gb/s, and defines services and protocol elements that enable the exchange of IEEE Std 802.3 format frames between stations in a subscriber access network. Clause 68 specifies a 10 Gb/s Physical Layer specification. Clause 69 through Clause 74 and associated annexes specify Ethernet operation over electrical backplanes at speeds of 1000 Mb/s and 10 Gb/s.

Section Six—Includes Clause 78 through Clause 95 and Annex 83A through Annex 93C. Clause 78 specifies Energy-Efficient Ethernet. Clause 79 specifies IEEE 802.3 Organizationally Specific Link Layer Discovery Protocol (LLDP) type, length, and value (TLV) information elements. Clause 80 through Clause 95 and associated annexes include general information on 40 Gb/s and 100 Gb/s operation as well as 40 Gb/s and 100 Gb/s Physical Layer specifications. Clause 90 specifies Ethernet support for time synchronization protocols.

Section Seven—Includes Clause 96 through Clause 115 and Annex 97A through Annex 115A. Clause 96 through Clause 98, Clause 104, and associated annexes, specify Physical Layers and optional features for 100 Mb/s and 1000 Mb/s operation over a single twisted pair. Clause 100 through Clause 103, as well as associated annexes, specify Physical Layers for the operation of the EPON protocol over coaxial distribution networks. Clause 105 through Clause 114 and associated annexes include general information on 25 Gb/s operation as well as 25 Gb/s Physical Layer specifications. Clause 99 specifies a MAC merge sublayer for the interspersing of express traffic. Clause 115 and its associated annex specify a Physical Layer for 1000 Mb/s operation over plastic optical fiber.

Section Eight—Includes Clause 116 through Clause 140 and Annex 119A through Annex 136D. Clause 116 through Clause 124 and associated annexes include general information on 200 Gb/s and 400 Gb/s operation as well as 200 Gb/s and 400 Gb/s Physical Layer specifications. Clause 125 includes general information on 2.5 Gb/s and 5 Gb/s operation. Clause 126 through Clause 130 and associated annexes include 2.5 Gb/s and 5 Gb/s Physical Layer specifications. Clause 131 provides general information on 50 Gb/s operation. Clause 132 through Clause 140 and associated annexes include 50 Gb/s Physical Layer specifications and additional 100 Gb/s, 200 Gb/s, and 400 Gb/s Physical Layer specifications.

Section Nine—Includes Clause 141 through Clause 160 and Annex 142A through Annex 154A. Clause 141 through Clause 144 and associated annexes specify symmetric and asymmetric operation of Ethernet passive optical networks over multiple 25 Gb/s channels. Clause 145 and associated annexes specify increased power delivery using all four pairs in the structured wiring plant. Clause 146 through Clause 149 and associated annexes specify Physical Layers for 10 Mb/s, 2.5 Gb/s, 5 Gb/s, and 10 Gb/s operation over a single balanced pair of conductors. Clause 150 and Clause 151 include additional 400 Gb/s Physical Layer specifications. Clause 153 and Clause 154 specify 100 Gb/s operation over DWDM channels. Clause 157 through Clause 160 include 10 Gb/s, 25 Gb/s, and 50 Gb/s bidirectional Physical Layer specifications.

## IEEE Std 802.3dd™-2022

Amendment 1—This amendment includes editorial and technical corrections, refinements, and clarifications to Clause 104, Power over Data Lines of Single Pair Ethernet, and related portions of the standard.

## IEEE Std 802.3cs™-2022

Amendment 2—This amendment to IEEE Std 802.3-2022 defines Super-PON optical subscriber access networks, in the family of Ethernet passive optical networks (EPONs). Super-PON has a reach of up to 50 km and up to 1024 ONUs over a point-to-multipoint passive optical distribution network (ODN)

through wavelength division multiplexing (WDM). A Super-PON ODN contains a passive wavelength router that determines the channels used by the ODN. This standard specifies the Super-PON Reconciliation Sublayer (RS), Physical Coding Sublayer (PCS), Physical Media Attachment (PMA) sublayer, and Physical Medium Dependent (PMD) sublayer at a MAC data rate of 10 Gb/s in the downstream direction and of 10 Gb/s or 2.5 Gb/s in the upstream direction.

#### IEEE Std 802.3db™-2022

Amendment 3—This amendment includes changes to IEEE Std 802.3-2022 and adds Clause 167. This amendment adds Physical Layer specifications and management parameters for 100 Gb/s, 200 Gb/s, and 400 Gb/s over one, two, and four pairs of multimode fiber based on 100 Gb/s optical signaling.

#### IEEE Std 802.3ck™-2022

Amendment 4—This amendment includes changes to IEEE Std 802.3-2022 and adds Clause 161 through Clause 163, Annex 120F, Annex 120G, Annex 162A through Annex 162D, Annex 163A, and Annex 163B. This amendment includes Physical Layer specifications and management parameters for 100 Gb/s, 200 Gb/s, and 400 Gb/s electrical interfaces based on 100 Gb/s signaling.

#### IEEE Std 802.3de™-2022

Amendment 5—This amendment includes changes to IEEE Std 802.3-2022 to add 10 Mb/s Single-Pair Ethernet point-to-point PHYs to the PHYs supporting the MAC Merge function and the Time Synchronization Service Interface (TSSI).

#### IEEE Std 802.3cx™-2023

Amendment 6—This amendment to IEEE Std 802.3-2022 modifies Clause 30, Clause 45, and Clause 90, and adds Annex 90A to enhance support for time synchronization protocols by providing options for sub-nanosecond reporting of the transmit and receive path data delays, selection of the data delay measurement point, and dynamic reporting of path data delay variation.

Two companion documents exist, IEEE Std 802.3.1 and IEEE Std 802.3.2. IEEE Std 802.3.1 describes Ethernet management information base (MIB) modules for use with the Simple Network Management Protocol (SNMP). IEEE Std 802.3.2 describes YANG data models for Ethernet. IEEE Std 802.3.1 and IEEE Std 802.3.2 are updated to add management capability for enhancements to IEEE Std 802.3 after approval of those enhancements.

IEEE Std 802.3 will continue to evolve. New Ethernet capabilities are anticipated to be added within the next few years as amendments to this standard.

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# IEEE Standard for Ethernet

## Amendment 6: Media Access Control (MAC) Service Interface and Management Parameters to Support Improved Precision Time Protocol (PTP) Timestamping Accuracy

(This amendment is based on IEEE Std 802.3™-2022 as amended by IEEE Std 802.3dd™-2022, IEEE Std 802.3cs™-2022, IEEE Std 802.3db™-2022, IEEE Std 802.3ck™-2022, and IEEE Std 802.3de™-2022.)

NOTE—The editing instructions contained in this amendment define how to merge the material contained therein into the existing base standard and its amendments to form the comprehensive standard.

The editing instructions are shown in ***bold italic***. Four editing instructions are used: change, delete, insert, and replace. ***Change*** is used to make corrections in existing text or tables. The editing instruction specifies the location of the change and describes what is being changed by using ~~strike through~~ (to remove old material) and underscore (to add new material). ***Delete*** removes existing material. ***Insert*** adds new material without disturbing the existing material. Deletions and insertions may require renumbering. If so, renumbering instructions are given in the editing instruction. ***Replace*** is used to make changes in figures or equations by removing the existing figure or equation and replacing it with a new one. Editing instructions, change markings, and this NOTE will not be carried over into future editions because the changes will be incorporated into the base standard.

Cross references that refer to clauses, tables, equations, or figures not covered by this amendment are highlighted in green.<sup>6</sup>

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<sup>6</sup> Notes in text, tables, and figures are given for information only and do not contain requirements needed to implement the standard.

## 1. Introduction

### 1.5 Abbreviations

*Insert the following new abbreviation into the list in 1.5, in alphanumeric order:*

|      |                              |
|------|------------------------------|
| DDMP | data delay measurement point |
|------|------------------------------|

## 30. Management

### 30.2 Managed objects

#### 30.2.5 Capabilities

Change Table 30–6 as follows:

**Table 30–6—TimeSync Capabilities**

|                                    |           |         |  |                                   |                                  |
|------------------------------------|-----------|---------|--|-----------------------------------|----------------------------------|
|                                    |           |         |  |                                   |                                  |
|                                    |           |         |  | Support for Time Sync (mandatory) | Support for Time Sync (optional) |
| <b>oTimeSync managed object</b>    |           |         |  |                                   |                                  |
| aTimeSyncCapabilityNsTX            | ATTRIBUTE | GET     |  | X                                 |                                  |
| aTimeSyncCapabilityNsRX            | ATTRIBUTE | GET     |  | X                                 |                                  |
| aTimeSyncDelayNsTXmax              | ATTRIBUTE | GET     |  | X                                 |                                  |
| aTimeSyncDelayNsTXmin              | ATTRIBUTE | GET     |  | X                                 |                                  |
| aTimeSyncDelayNsRXmax              | ATTRIBUTE | GET     |  | X                                 |                                  |
| aTimeSyncDelayNsRXmin              | ATTRIBUTE | GET     |  | X                                 |                                  |
| aTimeSyncCapabilitySubNsTX         | ATTRIBUTE | GET     |  |                                   | X                                |
| aTimeSyncCapabilitySubNsRX         | ATTRIBUTE | GET     |  |                                   | X                                |
| aTimeSyncDelaySubNsTXmax           | ATTRIBUTE | GET     |  |                                   | X                                |
| aTimeSyncDelaySubNsTXmin           | ATTRIBUTE | GET     |  |                                   | X                                |
| aTimeSyncDelaySubNsRXmax           | ATTRIBUTE | GET     |  |                                   | X                                |
| aTimeSyncDelaySubNsRXmin           | ATTRIBUTE | GET     |  |                                   | X                                |
| aTimeSyncCapabilityDdmp            | ATTRIBUTE | GET     |  |                                   | X                                |
| aTimeSyncSelectionDdmp             | ATTRIBUTE | GET-SET |  |                                   | X                                |
| aTimeSyncCapabilityMultiplePcsLane | ATTRIBUTE | GET     |  |                                   | X                                |
| aTimeSyncCapabilityDynamicPath-    | ATTRIBUTE | GET     |  |                                   | X                                |

### 30.13 Management for oTimeSync entity

#### 30.13.1 TimeSync entity managed object class

*Change the title and content of 30.13.1.1 as follows:*

##### 30.13.1.1 aTimeSyncCapabilityNsTX

ATTRIBUTE

APPROPRIATE SYNTAX:  
BOOLEAN

BEHAVIOUR DEFINED AS:

True if the TimeSync path data delay provisioning capability, with nanosecond resolution, is supported in the transmit path and false otherwise.

If a Clause 45 MDIO Interface to PMA/PMD, WIS, PCS, PHY XS, DTE XS, and/or TC is present, then the value stored in this attribute is equal to the logical OR operation over the values stored in the following instantiated MDIO registers (for each MMD, in case of multiple instances):

— For PMA/PMD: 1.1800.1, see 45.2.1.175

— For WIS: 2.1800.1, see 45.2.2.20

— For PCS: 3.1800.1, see 45.2.3.67

— For PHY XS: 4.1800.1, see 45.2.4.28

— For DTE XS: 5.1800.1, see 45.2.5.28

— For TC: ~~and~~ 6.1800.1, see 45.2.6.14;

(see 45.2.1.175, 45.2.2.20, 45.2.3.67, 45.2.4.28, 45.2.5.28, 45.2.6.14, respectively).

*Change the title and content of 30.13.1.2 as follows:*

##### 30.13.1.2 aTimeSyncCapabilityNsRX

ATTRIBUTE

APPROPRIATE SYNTAX:  
BOOLEAN

BEHAVIOUR DEFINED AS:

True if the TimeSync path data delay provisioning capability, with nanosecond resolution, is supported in the receive path and false otherwise.

If a Clause 45 MDIO Interface to PMA/PMD, WIS, PCS, PHY XS, DTE XS, and/or TC is present, then the value stored in this attribute is equal to the logical OR operation over the values stored in the following instantiated MDIO registers (for each MMD, in case of multiple instances):

— For PMA/PMD: 1.1800.0, see 45.2.1.175

— For WIS: 2.1800.0, see 45.2.2.20

— For PCS: 3.1800.0, see 45.2.3.67

— For PHY XS: 4.1800.0, see 45.2.4.28

— For DTE XS: 5.1800.0, see 45.2.5.28

— For TC: ~~and~~ 6.1800.0, see 45.2.6.14;

(see 45.2.1.175, 45.2.2.20, 45.2.3.67, 45.2.4.28, 45.2.5.28, 45.2.6.14, respectively).

*Change the title and content of 30.13.1.3 as follows:*

### **30.13.1.3 aTimeSyncDelayNsTXmax**

ATTRIBUTE

APPROPRIATE SYNTAX:

INTEGER

BEHAVIOUR DEFINED AS:

The integer nanosecond portion of the maximum transmit path data delay as specified in 90.7, expressed in units of ns.

If a Clause 45 MDIO Interface to PMA/PMD, WIS, PCS, PHY XS, DTE XS<sub>1</sub> and/or TC is present, then the value stored in this attribute represents the maximum transmit path data delay values, consisting of the sum of the values of the ~~registers~~ register sets in the instantiated sublayers (for each MMD, in case of multiple instances):

- ~~for~~ For PMA/PMD: 1.1801 and 1.1802, see 45.2.1.176
- ~~for~~ For WIS: 2.1801 and 2.1802, see 45.2.2.21
- ~~for~~ For PCS: 3.1801 and 3.1802, see 45.2.3.68
- ~~for~~ For PHY XS: 4.1801 and 4.1802, see 45.2.4.29
- ~~for~~ For DTE XS: 5.1801 and 5.1802, see 45.2.5.29
- ~~for~~ For TC: 6.1801 and 6.1802, see 45.2.6.15.;

*Change the title and content of 30.13.1.4 as follows:*

### **30.13.1.4 aTimeSyncDelayNsTXmin**

ATTRIBUTE

APPROPRIATE SYNTAX:

INTEGER

BEHAVIOUR DEFINED AS:

The integer nanosecond portion of the minimum transmit path data delay as specified in 90.7, expressed in units of ns.

If a Clause 45 MDIO Interface to PMA/PMD, WIS, PCS, PHY XS, DTE XS<sub>1</sub> and/or TC is present, then the value stored in this attribute represents the minimum transmit path data delay values, consisting of the sum of the values of the ~~registers~~ register sets in the instantiated sublayers (for each MMD, in case of multiple instances):

- ~~for~~ For PMA/PMD: 1.1803 and 1.1804, see 45.2.1.176
- ~~for~~ For WIS: 2.1803 and 2.1804, see 45.2.2.21
- ~~for~~ For PCS: 3.1803 and 3.1804, see 45.2.3.68
- ~~for~~ For PHY XS: 4.1803 and 4.1804, see 45.2.4.29
- ~~for~~ For DTE XS: 5.1803 and 5.1804, see 45.2.5.29
- ~~for~~ For TC: 6.1803 and 6.1804, see 45.2.6.15.;

*Change the title and content of 30.13.1.5 as follows:*

### **30.13.1.5 aTimeSyncDelayNsRXmax**

ATTRIBUTE

APPROPRIATE SYNTAX:

INTEGER

BEHAVIOUR DEFINED AS:

The integer nanosecond portion of the maximum receive path data delay as specified in 90.7, expressed in units of ns.

If a Clause 45 MDIO Interface to PMA/PMD, WIS, PCS, PHY XS, DTE XS, and/or TC is present, then the value stored in this attribute represents the maximum receive path data delay values, consisting of the sum of the values of the registers-register sets in the instantiated sublayers (for each MMD, in case of multiple instances):

- ~~for~~For PMA/PMD: 1.1805 and 1.1806, see 45.2.1.177
- ~~for~~For WIS: 2.1805 and 2.1806, see 45.2.2.22
- ~~for~~For PCS: 3.1805 and 3.1806, see 45.2.3.69
- ~~for~~For PHY XS: 4.1805 and 4.1806, see 45.2.4.30
- ~~for~~For DTE XS: 5.1805 and 5.1806, see 45.2.5.30
- ~~for~~For TC: 6.1805 and 6.1806, see 45.2.6.16.;

*Change the title and content of 30.13.1.6 as follows:*

### **30.13.1.6 aTimeSyncDelayNsRXmin**

ATTRIBUTE

APPROPRIATE SYNTAX:

INTEGER

BEHAVIOUR DEFINED AS:

The integer nanosecond portion of the minimum receive path data delay as specified in 90.7, expressed in units of ns.

If a Clause 45 MDIO Interface ~~to~~ to PMA/PMD, WIS, PCS, PHY XS, DTE XS, and/or TC is present, then the value stored in this attribute represents the minimum receive path data delay values, consisting of the sum of the values of the registers-register sets in the instantiated sublayers (for each MMD, in case of multiple instances):

- ~~for~~For PMA/PMD: 1.1807 and 1.1808, see 45.2.1.177
- ~~for~~For WIS: 2.1807 and 2.1808, see 45.2.2.22
- ~~for~~For PCS: 3.1807 and 3.1808, see 45.2.3.69
- ~~for~~For PHY XS: 4.1807 and 4.1808, see 45.2.4.30
- ~~for~~For DTE XS: 5.1807 and 5.1808, see 45.2.5.30
- ~~for~~For TC: 6.1807 and 6.1808, see 45.2.6.16.;

*Insert 30.13.1.7 to 30.13.1.16 after 30.13.1.6 as follows:*

### **30.13.1.7 aTimeSyncCapabilitySubNsTX**

ATTRIBUTE

APPROPRIATE SYNTAX:

BOOLEAN

BEHAVIOUR DEFINED AS:

True if the TimeSync path data delay provisioning capability, with sub-nanosecond resolution, is supported in the transmit path and false otherwise.

If a Clause 45 MDIO Interface to PMA/PMD, WIS, PCS, PHY XS, DTE XS, and/or TC is present, then the value stored in this attribute is equal to the logical OR operation over the values stored in the following instantiated MDIO registers (for each MMD, in case of multiple instances):

- For PMA/PMD: 1.1800.3, see 45.2.1.175
- For WIS: 2.1800.3, see 45.2.2.20
- For PCS: 3.1800.3, see 45.2.3.67
- For PHY XS: 4.1800.3, see 45.2.4.28
- For DTE XS: 5.1800.3, see 45.2.5.28
- For TC: 6.1800.3, see 45.2.6.14.;

### **30.13.1.8 aTimeSyncCapabilitySubNsRX**

ATTRIBUTE

APPROPRIATE SYNTAX:

BOOLEAN

BEHAVIOUR DEFINED AS:

True if the TimeSync path data delay provisioning capability, with sub-nanosecond resolution, is supported in the receive path and false otherwise.

If a Clause 45 MDIO Interface to PMA/PMD, WIS, PCS, PHY XS, DTE XS, and/or TC is present, then the value stored in this attribute is equal to the logical OR operation over the values stored in the following instantiated MDIO registers (for each MMD, in case of multiple instances):

- For PMA/PMD: 1.1800.2, see 45.2.1.175
- For WIS: 2.1800.2, see 45.2.2.20
- For PCS: 3.1800.2, see 45.2.3.67
- For PHY XS: 4.1800.2, see 45.2.4.28
- For DTE XS: 5.1800.2, see 45.2.5.28
- For TC: 6.1800.2, see 45.2.6.14.;

### **30.13.1.9 aTimeSyncDelaySubNsTXmax**

ATTRIBUTE

APPROPRIATE SYNTAX:

INTEGER

BEHAVIOUR DEFINED AS:

The sub-nanosecond portion of the maximum transmit path data delay as specified in 90.7, expressed in units of  $2^{-16}$  ns.

If a Clause 45 MDIO Interface to PMA/PMD, WIS, PCS, PHY XS, DTE XS, and/or TC is present,

then the value stored in this attribute represents the maximum transmit path data delay values, consisting of the sum of the values of the register sets in the instantiated sublayers (for each MMD, in case of multiple instances):

- For PMA/PMD: 1.1809, see 45.2.1.176
- For WIS: 2.1809, see 45.2.2.21
- For PCS: 3.1809, see 45.2.3.68
- For PHY XS: 4.1809, see 45.2.4.29
- For DTE XS: 5.1809, see 45.2.5.29
- For TC: 6.1809, see 45.2.6.15.;

### 30.13.1.10 aTimeSyncDelaySubNsTXmin

ATTRIBUTE

APPROPRIATE SYNTAX:

INTEGER

BEHAVIOUR DEFINED AS:

The sub-nanosecond portion of the minimum transmit path data delay as specified in 90.7, expressed in units of  $2^{-16}$  ns.

If a Clause 45 MDIO Interface to PMA/PMD, WIS, PCS, PHY XS, DTE XS, and/or TC is present, then the value stored in this attribute represents the minimum transmit path data delay values, consisting of the sum of the values of the register sets in the instantiated sublayers (for each MMD, in case of multiple instances):

- For PMA/PMD: 1.1810, see 45.2.1.176
- For WIS: 2.1810, see 45.2.2.21
- For PCS: 3.1810, see 45.2.3.68
- For PHY XS: 4.1810, see 45.2.4.29
- For DTE XS: 5.1810, see 45.2.5.29
- For TC: 6.1810, see 45.2.6.15.;

### 30.13.1.11 aTimeSyncDelaySubNsRXmax

ATTRIBUTE

APPROPRIATE SYNTAX:

INTEGER

BEHAVIOUR DEFINED AS:

The sub-nanosecond portion of the maximum receive path data delay as specified in 90.7, expressed in units of  $2^{-16}$  ns.

If a Clause 45 MDIO Interface to PMA/PMD, WIS, PCS, PHY XS, DTE XS, and/or TC is present, then the value stored in this attribute represents the maximum receive path data delay values, consisting of the sum of the values of the register sets in the instantiated sublayers (for each MMD, in case of multiple instances):

- For PMA/PMD: 1.1811, see 45.2.1.177
- For WIS: 2.1811, see 45.2.2.22
- For PCS: 3.1811, see 45.2.3.69
- For PHY XS: 4.1811, see 45.2.4.30
- For DTE XS: 5.1811, see 45.2.5.30
- For TC: 6.1811, see 45.2.6.16.;

### 30.13.1.12 aTimeSyncDelaySubNsRXmin

ATTRIBUTE

APPROPRIATE SYNTAX:  
INTEGER

BEHAVIOUR DEFINED AS:

The sub-nanosecond portion of the minimum receive path data delay as specified in 90.7, expressed in units of  $2^{-16}$  ns.

If a Clause 45 MDIO Interface to PMA/PMD, WIS, PCS, PHY XS, DTE XS, and/or TC is present, then the value stored in this attribute represents the minimum receive path data delay values, consisting of the sum of the values of the register sets in the instantiated sublayers (for each MMD, in case of multiple instances):

- For PMA/PMD: 1.1812, see 45.2.1.177
- For WIS: 2.1812, see 45.2.2.22
- For PCS: 3.1812, see 45.2.3.69
- For PHY XS: 4.1812, see 45.2.4.30
- For DTE XS: 5.1812, see 45.2.5.30
- For TC: 6.1812, see 45.2.6.16.;

### 30.13.1.13 aTimeSyncCapabilityDdmp

ATTRIBUTE

APPROPRIATE SYNTAX:

A SEQUENCE that meets the requirements of the description below:

|                  |                                                                                      |
|------------------|--------------------------------------------------------------------------------------|
| sfd              | Capable of using the beginning of the SFD as the data delay measurement point (DDMP) |
| symbol-after-sfd | Capable of using the beginning of the first symbol after the SFD as the DDMP         |

BEHAVIOUR DEFINED AS:

A read-only list of the supported DDMP options.

If a Clause 45 MDIO Interface to the PCS and/or DTE XS is present, then the value stored in this attribute is calculated based on the values stored in the following instantiated MDIO registers (for each MMD, in case of multiple instances).

The value of 'sfd' indicates that register bits 3.1800.13 and 5.1800.13 (see 45.2.3.67 and 45.2.5.28) are both set to one or that all register bits 3.1800.12, 3.1800.13, 5.1800.12, and 5.1800.13 are set to zero.

The value of 'symbol-after-sfd' indicates that register bits 3.1800.12 and 5.1800.12 (see 45.2.3.67 and 45.2.5.28) are both set to one.;

### 30.13.1.14 aTimeSyncSelectionDdmp

ATTRIBUTE

APPROPRIATE SYNTAX:

An ENUMERATED VALUE that takes one of the following entries:

|                  |                                                                     |
|------------------|---------------------------------------------------------------------|
| sfd              | The beginning of the SFD is used as the DDMP                        |
| symbol-after-sfd | The beginning of the first symbol after the SFD is used as the DDMP |

BEHAVIOUR DEFINED AS:

The SET operation configures the DDMP.

If a Clause 45 MDIO Interface to the PCS and/or DTE XS is present, then a SET operation

configures the DDMP that is used by the PCS and/or the DTE XS. The SET operation configures the following instantiated MDIO registers (for each MMD, in case of multiple instances). The value of 'sfd' configures register bits 3.1813.13 and/or 5.1813.13 (see 45.2.3.69a and 45.2.5.31) to zero.

The value of 'symbol-after-sfd' configures register bits 3.1813.13 and/or 5.1813.13 (see 45.2.3.69a and 45.2.5.31) to one.

The GET operation shows the configured DDMP.

If a Clause 45 MDIO Interface to the PCS and/or DTE XS is present, then the value returned by this attribute is calculated as follows, based on values stored in the following instantiated MDIO registers (for each MMD, in case of multiple instances).

The value of 'sfd' indicates that register bits 3.1813.13 and 5.1813.13 (see 45.2.3.69a and 45.2.5.31) are both set to zero.

The value of 'symbol-after-sfd' indicates that register bits 3.1813.13 and 5.1813.13 (see 45.2.3.69a and 45.2.5.31) are both set to one.

If both PCS and DTE XS functions are present, register bits 3.1813.13 and 5.1813.13 (see 45.2.3.69a.1 and 45.2.5.31.1) are expected to be set to the same value, assuming that this setting is supported by the data delay measurement point abilities in both the PCS and DTE XS TimeSync capability registers (see 45.2.3.67 and 45.2.5.28). If different values are used for the PCS and the DTE XS, error might be added to the measured path data delay.;

### **30.13.1.15 aTimeSyncCapabilityMultiplePcsLane**

ATTRIBUTE

APPROPRIATE SYNTAX:

BOOLEAN

BEHAVIOUR DEFINED AS:

True if the TimeSync multiple PCS lane path data delay reporting capability is supported and false otherwise.

If a Clause 45 MDIO Interface to the PCS is present, then the value stored in this attribute is equal to the value stored in the following instantiated MDIO register (for each MMD, in case of multiple instances):

— For PCS: 3.1800.11, see 45.2.3.67.;

### **30.13.1.16 aTimeSyncCapabilityDynamicPathDataDelay**

ATTRIBUTE

APPROPRIATE SYNTAX:

BOOLEAN

BEHAVIOUR DEFINED AS:

True if the TimeSync PCS dynamic path data delay capability is supported and false otherwise.

If a Clause 45 MDIO Interface to the PCS is present, then the value stored in this attribute is equal to the value stored in the following instantiated MDIO register (for each MMD, in case of multiple instances):

— For PCS: 3.1800.10, see 45.2.3.67.;

## 45. Management Data Input/Output (MDIO) Interface

### 45.2 MDIO Interface registers

#### 45.2.1 PMA/PMD registers

*Change the rows for register addresses 1.1801 to 1.1899 in Table 45–3 as follows (unchanged rows not shown):*

**Table 45–3—PMA/PMD registers**

| Register address         | Register name                                              | Subclause         |
|--------------------------|------------------------------------------------------------|-------------------|
| ...                      |                                                            |                   |
| 1.1801 through 1.1804    | TimeSync PMA/PMD transmit path data delay <u>in ns</u>     | 45.2.1.176        |
| 1.1805 through 1.1808    | TimeSync PMA/PMD receive path data delay <u>in ns</u>      | 45.2.1.177        |
| <u>1.1809 and 1.1810</u> | <u>TimeSync PMA/PMD transmit path data delay in sub-ns</u> | <u>45.2.1.176</u> |
| <u>1.1811 and 1.1812</u> | <u>TimeSync PMA/PMD receive path data delay in sub-ns</u>  | <u>45.2.1.177</u> |
| 1.1813 through 1.1899    | Reserved                                                   |                   |
| ...                      |                                                            |                   |

#### 45.2.1.175 TimeSync PMA/PMD capability (Register 1.1800)

*Change the text of 45.2.1.175 as follows:*

The TimeSync PMA/PMD capability register (see Table 45–139) indicates the capability of the PMA/PMD to report the transmit and receive path data delay, stored in registers 1.1801 and 1.1805 through 1.1808 respectively (with ns resolution in registers 1.1801 through 1.1804 and, separately, with sub-ns resolution in registers 1.1809 and 1.1810) and receive path data delay (with ns resolution in registers 1.1805 through 1.1808 and, separately, with sub-ns resolution in registers 1.1811 and 1.1812). See 90.7.

*Change Table 45–139 as follows:*

**Table 45–139—TimeSync PMA/PMD capability**

| Bit(s)          | Name                                                        | Description                                                                                                                                                                                                                     | R/W <sup>a</sup> |
|-----------------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.1800.15:24    | Reserved                                                    | Value always 0                                                                                                                                                                                                                  | RO               |
| <u>1.1800.3</u> | <u>TimeSync transmit path data delay ability, in sub-ns</u> | <u>1 = PMA/PMD provides information on transmit path data delay with sub-ns resolution in registers 1.1809 and 1.1810</u><br><u>0 = PMA/PMD does not provide information on transmit path data delay with sub-ns resolution</u> | <u>RO</u>        |
| <u>1.1800.2</u> | <u>TimeSync receive path data delay ability, in sub-ns</u>  | <u>1 = PMA/PMD provides information on receive path data delay with sub-ns resolution in registers 1.1811 and 1.1812</u><br><u>0 = PMA/PMD does not provide information on receive path data delay with sub-ns resolution</u>   | <u>RO</u>        |

**Table 45–139—TimeSync PMA/PMD capability (*continued*)**

| Bit(s)   | Name                                                    | Description                                                                                                                                                                                                                 | R/W <sup>a</sup> |
|----------|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.1800.1 | TimeSync transmit path data delay <u>ability, in ns</u> | 1 = PMA/PMD provides information on transmit path data delay <u>with ns resolution</u> in registers 1.1801 through 1.1804<br>0 = PMA/PMD does not provide information on transmit path data delay <u>with ns resolution</u> | RO               |
| 1.1800.0 | TimeSync receive path data delay <u>ability, in ns</u>  | 1 = PMA/PMD provides information on receive path data delay <u>with ns resolution</u> in registers 1.1805 through 1.1808<br>0 = PMA/PMD does not provide information on receive path data delay <u>with ns resolution</u>   | RO               |

<sup>a</sup>RO = Read only.

*Change the title and text of 45.2.1.176 as follows:*

**45.2.1.176 TimeSync PMA/PMD transmit path data delay (Registers 1.1801, 1.1802, 1.1803, 1.1804, 1.1809, and 1.1810)**

~~The TimeSync PMA/PMD transmit path data delay register contains the maximum (registers 1.1801, 1.1802, see Table 45–140) and minimum (registers 1.1803, 1.1804, see Table 45–140) values of the transmit path data delay. The transmit path data delay is expressed in units of ns. The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 1.1 (see 45.2.1.2.4).~~

The TimeSync PMA/PMD transmit path data delay registers contain the maximum and minimum values of the PMA/PMD transmit path data delay (see 90.7). The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 1.1 (see 45.2.1.2.4), and when the corresponding capability register bit (see Table 45–139) is asserted.

The maximum PMA/PMD transmit path data delay value is provided in two components. The first component (registers 1.1801 and 1.1802, see Table 45–140) provides the integer nanoseconds portion of the maximum PMA/PMD transmit path data delay, in units of nanoseconds. The second component (register 1.1809, see Table 45–140) provides the fractional nanoseconds portion of the maximum PMA/PMD transmit path data delay, in units of  $2^{-16}$  ns. If both components are valid (registers 1.1800.1 and 1.1800.3, see Table 45–139), the maximum PMA/PMD transmit path data delay is the sum of the values from these two components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the maximum PMA/PMD transmit path data delay.

The minimum PMA/PMD transmit path data delay value is provided in two components. The first component (registers 1.1803 and 1.1804, see Table 45–140) provides the integer nanoseconds portion of the minimum PMA/PMD transmit path data delay, in units of nanoseconds. The second component (register 1.1810, see Table 45–140) provides the fractional nanoseconds portion of the minimum PMA/PMD transmit path data delay, in units of  $2^{-16}$  ns. If both components are valid (registers 1.1800.1 and 1.1800.3, see Table 45–139), the minimum PMA/PMD transmit path data delay is the sum of the values from these two components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the minimum PMA/PMD transmit path data delay.

*Change Table 45–140 as follows:*

**Table 45–140—TimeSync PMA/PMD transmit path data delay register**

| Bit(s)             | Name                                                      | Description                                                                   | R/W <sup>a</sup> |
|--------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------|------------------|
| 1.1801.15:0        | Maximum PMA/PMD transmit path data delay in ns, lower     | <u>PMA/PMD_delay_TX_max [15:0]</u><br><u>PMA/PMD_delay_ns_TX_max [15:0]</u>   | RO, MW           |
| 1.1802.15:0        | Maximum PMA/PMD transmit path data delay in ns, upper     | <u>PMA/PMD_delay_TX_max [31:16]</u><br><u>PMA/PMD_delay_ns_TX_max [31:16]</u> | RO, MW           |
| 1.1803.15:0        | Minimum PMA/PMD transmit path data delay in ns, lower     | <u>PMA/PMD_delay_TX_min [15:0]</u><br><u>PMA/PMD_delay_ns_TX_min [15:0]</u>   | RO, MW           |
| 1.1804.15:0        | Minimum PMA/PMD transmit path data delay in ns, upper     | <u>PMA/PMD_delay_TX_min [31:16]</u><br><u>PMA/PMD_delay_ns_TX_min [31:16]</u> | RO, MW           |
| <u>1.1809.15:0</u> | <u>Maximum PMA/PMD transmit path data delay in sub-ns</u> | <u>PMA/PMD_delay_subns_TX_max [15:0]</u>                                      | <u>RO</u>        |
| <u>1.1810.15:0</u> | <u>Minimum PMA/PMD transmit path data delay in sub-ns</u> | <u>PMA/PMD_delay_subns_TX_min [15:0]</u>                                      | <u>RO</u>        |

<sup>a</sup>RO = Read only, MW = Multi-word.

*Change the title and text of 45.2.1.177 as follows:*

**45.2.1.177 TimeSync PMA/PMD receive path data delay (Registers 1.1805, 1.1806, 1.1807, 1.1808, 1.1811, and 1.1812)**

The TimeSync PMA/PMD receive path data delay register contains the maximum (registers 1.1805, 1.1806, see Table 45–141) and minimum (registers 1.1807, 1.1808, see Table 45–141) values of the receive path data delay. The receive path data delay is expressed in units of ns. The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 1.1 (see 45.2.1.2.4).

The TimeSync PMA/PMD receive path data delay registers contain the maximum and minimum values of the PMA/PMD receive path data delay (see 90.7). The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 1.1 (see 45.2.1.2.4), and when the corresponding capability register bit (see Table 45–139) is asserted.

The maximum PMA/PMD receive path data delay value is provided in two components. The first component (registers 1.1805 and 1.1806, see Table 45–141) provides the integer nanoseconds portion of the maximum PMA/PMD receive path data delay, in units of nanoseconds. The second component (register 1.1811, see Table 45–141) provides the fractional nanoseconds portion of the maximum PMA/PMD receive path data delay, in units of  $2^{-16}$  ns. If both components are valid (registers 1.1800.0 and 1.1800.2, see Table 45–139), the maximum PMA/PMD receive path data delay is the sum of the values from these two components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the maximum PMA/PMD receive path data delay.

The minimum PMA/PMD receive path data delay value is provided in two components. The first component (registers 1.1807 and 1.1808, see Table 45–141) provides the integer nanoseconds portion of the minimum PMA/PMD receive path data delay, in units of nanoseconds. The second component (register 1.1812, see Table 45–141) provides the fractional nanoseconds portion of the minimum PMA/PMD receive path data delay, in units of  $2^{-16}$  ns. If both components are valid (registers 1.1800.0 and 1.1800.2, see Table 45–139), the minimum PMA/PMD receive path data delay is the sum of the values from these two

components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the minimum PMA/PMD receive path data delay.

*Change Table 45–141 as follows:*

**Table 45–141—TimeSync PMA/PMD receive path data delay register**

| Bit(s)             | Name                                                         | Description                                                                       | R/W <sup>a</sup> |
|--------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------|
| 1.1805.15:0        | Maximum PMA/PMD receive path data delay <u>in ns</u> , lower | <del>PMA/PMD_delay_RX_max [15:0]</del><br><u>PMA/PMD_delay_ns_RX_max [15:0]</u>   | RO, MW           |
| 1.1806.15:0        | Maximum PMA/PMD receive path data delay <u>in ns</u> , upper | <del>PMA/PMD_delay_RX_max [31:16]</del><br><u>PMA/PMD_delay_ns_RX_max [31:16]</u> | RO, MW           |
| 1.1807.15:0        | Minimum PMA/PMD receive path data delay <u>in ns</u> , lower | <del>PMA/PMD_delay_RX_min [15:0]</del><br><u>PMA/PMD_delay_ns_RX_min [15:0]</u>   | RO, MW           |
| 1.1808.15:0        | Minimum PMA/PMD receive path data delay <u>in ns</u> , upper | <del>PMA/PMD_delay_RX_min [31:16]</del><br><u>PMA/PMD_delay_ns_RX_min [31:16]</u> | RO, MW           |
| <u>1.1811.15:0</u> | <u>Maximum PMA/PMD receive path data delay in sub-ns</u>     | <u>PMA/PMD_delay_subns_RX_max [15:0]</u>                                          | <u>RO</u>        |
| <u>1.1812.15:0</u> | <u>Minimum PMA/PMD receive path data delay in sub-ns</u>     | <u>PMA/PMD_delay_subns_RX_min [15:0]</u>                                          | <u>RO</u>        |

<sup>a</sup>RO = Read only, MW = Multi-word.

## 45.2.2 WIS registers

*Change the rows for register addresses 2.1801 to 2.32767 in Table 45–213 as follows (unchanged rows not shown):*

**Table 45–213—WIS registers**

| Register address         | Register name                                          | Subclause        |
|--------------------------|--------------------------------------------------------|------------------|
| ...                      |                                                        |                  |
| 2.1801 through 2.1804    | TimeSync WIS transmit path data delay <u>in ns</u>     | 45.2.2.21        |
| 2.1805 through 2.1808    | TimeSync WIS receive path data delay <u>in ns</u>      | 45.2.2.22        |
| <u>2.1809 and 2.1810</u> | <u>TimeSync WIS transmit path data delay in sub-ns</u> | <u>45.2.2.21</u> |
| <u>2.1811 and 2.1812</u> | <u>TimeSync WIS receive path data delay in sub-ns</u>  | <u>45.2.2.22</u> |
| 2.1813 through 2.32767   | Reserved                                               |                  |
| ...                      |                                                        |                  |

#### 45.2.2.20 TimeSync WIS capability (Register 2.1800)

*Change the text of 45.2.2.20 as follows:*

The TimeSync WIS capability register (see Table 45–230) indicates the capability of the WIS to report the transmit and receive path data delay, stored in registers 2.1801 through 2.1804 and 2.1805 through 2.1808, respectively (with ns resolution in registers 2.1801 through 2.1804 and, separately, with sub-ns resolution in registers 2.1809 and 2.1810) and receive path data delay (with ns resolution in registers 2.1805 through 2.1808 and, separately, with sub-ns resolution in registers 2.1811 and 2.1812). See 90.7.

*Change Table 45–230 as follows:*

**Table 45–230—TimeSync WIS capability**

| Bit(s)                  | Name                                                        | Description                                                                                                                                                                                                             | R/W <sup>a</sup> |
|-------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 2.1800.15:4<br><u>2</u> | Reserved                                                    | Value always 0                                                                                                                                                                                                          | RO               |
| <u>2.1800.3</u>         | <u>TimeSync transmit path data delay ability, in sub-ns</u> | <u>1 = WIS provides information on transmit path data delay with sub-ns resolution in registers 2.1809 and 2.1810</u><br><u>0 = WIS does not provide information on transmit path data delay with sub-ns resolution</u> | <u>RO</u>        |
| <u>2.1800.2</u>         | <u>TimeSync receive path data delay ability, in sub-ns</u>  | <u>1 = WIS provides information on receive path data delay with sub-ns resolution in registers 2.1811 and 2.1812</u><br><u>0 = WIS does not provide information on receive path data delay with sub-ns resolution</u>   | <u>RO</u>        |
| 2.1800.1                | TimeSync transmit path data delay, <u>in ns</u>             | 1 = WIS provides information on transmit path data delay <u>with ns resolution</u> in registers 2.1801 through 2.1804<br>0 = WIS does not provide information on transmit path data delay <u>with ns resolution</u>     | RO               |
| 2.1800.0                | TimeSync receive path data delay, <u>in ns</u>              | 1 = WIS provides information on receive path data delay <u>with ns resolution</u> in registers 2.1805 through 2.1808<br>0 = WIS does not provide information on receive path data delay <u>with ns resolution</u>       | RO               |

<sup>a</sup>RO = Read only

*Change the title and text of 45.2.2.21 as follows:*

#### 45.2.2.21 TimeSync WIS transmit path data delay (Registers 2.1801, 2.1802, 2.1803, 2.1804, 2.1809, and 2.1810)

The TimeSync WIS transmit path data delay register contains the maximum (registers 2.1801, 2.1802, see Table 45–231) and minimum (registers 2.1803, 2.1804, see Table 45–231) values of the transmit path data delay. The transmit path data delay is expressed in units of ns. The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 1.1 (see 45.2.1.2.4).

The TimeSync WIS transmit path data delay registers contain the maximum and minimum values of the WIS transmit path data delay (see 90.7). The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 2.1 (see 45.2.2.2.2), and when the corresponding capability register bit (see Table 45–230) is asserted.

The maximum WIS transmit path data delay value is provided in two components. The first component (registers 2.1801 and 2.1802, see Table 45–231) provides the integer nanoseconds portion of the maximum WIS transmit path data delay, in units of nanoseconds. The second component (register 2.1809, see Table 45–231) provides the fractional nanoseconds portion of the maximum WIS transmit path data delay, in units of  $2^{-16}$  ns. If both components are valid (registers 2.1800.1 and 2.1800.3, see Table 45–230), the maximum WIS transmit path data delay is the sum of the values from these two components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the maximum WIS transmit path data delay.

The minimum WIS transmit path data delay value is provided in two components. The first component (registers 2.1803 and 2.1804, see Table 45–231) provides the integer nanoseconds portion of the minimum WIS transmit path data delay, in units of nanoseconds. The second component (register 2.1810, see Table 45–231) provides the fractional nanoseconds portion of the minimum WIS transmit path data delay, in units of  $2^{-16}$  ns. If both components are valid (registers 2.1800.1 and 2.1800.3, see Table 45–230), the minimum WIS transmit path data delay is the sum of the values from these two components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the minimum WIS transmit path data delay.

*Change Table 45–231 as follows:*

**Table 45–231—TimeSync WIS transmit path data delay register**

| Bit(s)             | Name                                                      | Description                                                               | R/W <sup>a</sup> |
|--------------------|-----------------------------------------------------------|---------------------------------------------------------------------------|------------------|
| 2.1801.15:0        | Maximum WIS transmit path data delay <u>in ns</u> , lower | <del>WIS_delay_TX_max [15:0]</del><br><u>WIS_delay_ns_TX_max [15:0]</u>   | RO, MW           |
| 2.1802.15:0        | Maximum WIS transmit path data delay <u>in ns</u> , upper | <del>WIS_delay_TX_max [31:16]</del><br><u>WIS_delay_ns_TX_max [31:16]</u> | RO, MW           |
| 2.1803.15:0        | Minimum WIS transmit path data delay <u>in ns</u> , lower | <del>WIS_delay_TX_min [15:0]</del><br><u>WIS_delay_ns_TX_min [15:0]</u>   | RO, MW           |
| 2.1804.15:0        | Minimum WIS transmit path data delay <u>in ns</u> , upper | <del>WIS_delay_TX_min [31:16]</del><br><u>WIS_delay_ns_TX_min [31:16]</u> | RO, MW           |
| <u>2.1809.15:0</u> | <u>Maximum WIS transmit path data delay in sub-ns</u>     | <u>WIS_delay_subns_TX_max [15:0]</u>                                      | <u>RO</u>        |
| <u>2.1810.15:0</u> | <u>Minimum WIS transmit path data delay in sub-ns</u>     | <u>WIS_delay_subns_TX_min [15:0]</u>                                      | <u>RO</u>        |

<sup>a</sup>RO = Read only, MW = Multi-word

*Change the title and text of 45.2.2.22 as follows:*

#### **45.2.2.22 TimeSync WIS receive path data delay (Registers 2.1805, 2.1806, 2.1807, 2.1808, 2.1811, and 2.1812)**

The TimeSync WIS receive path data delay register contains the maximum (registers 2.1805, 2.1806, see Table 45–232) and minimum (registers 2.1807, 2.1808, see Table 45–232) values of the receive path data delay. The receive path data delay is expressed in units of ns. The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 2.1 (see 45.2.2.2.2).

The TimeSync WIS receive path data delay registers contain the maximum and minimum values of the WIS receive path data delay (see 90.7). The values contained in these registers are valid when the link is

established, as indicated by bit 2 in register 2.1 (see 45.2.2.2.2), and when the corresponding capability register bit (see Table 45–230) is asserted.

The maximum WIS receive path data delay value is provided in two components. The first component (registers 2.1805 and 2.1806, see Table 45–232) provides the integer nanoseconds portion of the maximum WIS receive path data delay, in units of nanoseconds. The second component (register 2.1811, see Table 45–232) provides the fractional nanoseconds portion of the maximum WIS receive path data delay, in units of  $2^{-16}$  ns. If both components are valid (registers 2.1800.0 and 2.1800.2, see Table 45–230), the maximum WIS receive path data delay is the sum of the values from these two components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the maximum WIS receive path data delay.

The minimum WIS receive path data delay value is provided in two components. The first component (registers 2.1807 and 2.1808, see Table 45–232) provides the integer nanoseconds portion of the minimum WIS receive path data delay, in units of nanoseconds. The second component (register 2.1812, see Table 45–232) provides the fractional nanoseconds portion of the minimum WIS receive path data delay, in units of  $2^{-16}$  ns. If both components are valid (registers 2.1800.0 and 2.1800.2, see Table 45–230), the minimum WIS receive path data delay is the sum of the values from these two components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the minimum WIS receive path data delay.

*Change Table 45–232 as follows:*

**Table 45–232—TimeSync WIS receive path data delay register**

| Bit(s)             | Name                                                     | Description                                                               | R/W <sup>a</sup> |
|--------------------|----------------------------------------------------------|---------------------------------------------------------------------------|------------------|
| 2.1805.15:0        | Maximum WIS receive path data delay <u>in ns</u> , lower | <del>WIS_delay_RX_max</del> [15:0]<br><u>WIS_delay_ns_RX_max</u> [15:0]   | RO, MW           |
| 2.1806.15:0        | Maximum WIS receive path data delay <u>in ns</u> , upper | <del>WIS_delay_RX_max</del> [31:16]<br><u>WIS_delay_ns_RX_max</u> [31:16] | RO, MW           |
| 2.1807.15:0        | Minimum WIS receive path data delay <u>in ns</u> , lower | <del>WIS_delay_RX_min</del> [15:0]<br><u>WIS_delay_ns_RX_min</u> [15:0]   | RO, MW           |
| 2.1808.15:0        | Minimum WIS receive path data delay <u>in ns</u> , upper | <del>WIS_delay_RX_min</del> [31:16]<br><u>WIS_delay_ns_RX_min</u> [31:16] | RO, MW           |
| <u>2.1811.15:0</u> | <u>Maximum WIS receive path data delay in sub-ns</u>     | <u>WIS_delay_subns_RX_max</u> [15:0]                                      | <u>RO</u>        |
| <u>2.1812.15:0</u> | <u>Minimum WIS receive path data delay in sub-ns</u>     | <u>WIS_delay_subns_RX_min</u> [15:0]                                      | <u>RO</u>        |

<sup>a</sup>RO = Read only, MW = Multi-word

### 45.2.3 PCS registers

*Change the rows for register addresses 3.1801 to 3.2277 in Table 45–233 as follows (unchanged rows not shown):*

**Table 45–233—PCS registers**

| Register address         | Register name                                          | Subclause         |
|--------------------------|--------------------------------------------------------|-------------------|
| ...                      |                                                        |                   |
| 3.1801 through 3.1804    | TimeSync PCS transmit path data delay <u>in ns</u>     | 45.2.3.68         |
| 3.1805 through 3.1808    | TimeSync PCS receive path data delay <u>in ns</u>      | 45.2.3.69         |
| <u>3.1809 and 3.1810</u> | <u>TimeSync PCS transmit path data delay in sub-ns</u> | <u>45.2.3.68</u>  |
| <u>3.1811 and 3.1812</u> | <u>TimeSync PCS receive path data delay in sub-ns</u>  | <u>45.2.3.69</u>  |
| <u>3.1813</u>            | <u>TimeSync PCS configuration</u>                      | <u>45.2.3.69a</u> |
| 3.1814 through 3.2277    | Reserved                                               |                   |
| ...                      |                                                        |                   |

#### 45.2.3.67 TimeSync PCS capability (Register 3.1800)

*Change the text of 45.2.3.67 as follows:*

The TimeSync PCS capability register (see Table 45–293) indicates the capability of the PCS to report the transmit and receive data delay, stored in registers 3.1801 through 3.1804 and 3.1805 through 3.1808, respectively.

This register is used to indicate the capability of the PCS to provide transmit and receive path data delay information in support of a TimeSync Client. See 90.7. The assignment of bits in the TimeSync PCS capability register is shown in Table 45–293.

*Change Table 45–293 as follows:*

**Table 45–293—TimeSync PCS capability**

| Bit(s)              | Name                                        | Description                                                                                                                                                                                                                                                                                                                                                       | R/W <sup>a</sup> |
|---------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <u>3.1800.15:14</u> | <u>Reserved</u>                             | <u>Value always 0</u>                                                                                                                                                                                                                                                                                                                                             | <u>RO</u>        |
| <u>3.1800.13:12</u> | <u>Data delay measurement point ability</u> | <u>x0 = PCS supports only the beginning of the SFD as the data delay measurement point</u><br><u>01 = PCS supports only the beginning of the first symbol after the SFD as the data delay measurement point</u><br><u>11 = PCS supports both the beginning of the SFD and the beginning of the first symbol after the SFD as the data delay measurement point</u> | <u>RO</u>        |
| <u>3.1800.11</u>    | <u>Multilane ability</u>                    | <u>0 = PCS does not support the reporting of multiple PCS lane path data delay using the method recommended in 90.7.1 and 90A.4</u><br><u>1 = PCS supports the reporting of multiple PCS lane path data delay using the method recommended in 90.7.1 and 90A.4</u>                                                                                                | <u>RO</u>        |

**Table 45–293—TimeSync PCS capability (*continued*)**

| Bit(s)                                      | Name                                                        | Description                                                                                                                                                                                                             | R/W <sup>a</sup> |
|---------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <u>3.1800.10</u>                            | <u>PCS dynamic path data delay ability</u>                  | <u>0 = PCS does not support PCS dynamic path data delay indication capability</u><br><u>1 = PCS supports PCS dynamic path data delay indication capability</u>                                                          | <u>RO</u>        |
| <del>3.1800.15:2</del><br><u>3.1800.9:4</u> | Reserved                                                    | Value always 0                                                                                                                                                                                                          | RO               |
| <u>3.1800.3</u>                             | <u>TimeSync transmit path data delay ability, in sub-ns</u> | <u>1 = PCS provides information on transmit path data delay with sub-ns resolution in registers 3.1809 and 3.1810</u><br><u>0 = PCS does not provide information on transmit path data delay with sub-ns resolution</u> | <u>RO</u>        |
| <u>3.1800.2</u>                             | <u>TimeSync receive path data delay ability, in sub-ns</u>  | <u>1 = PCS provides information on receive path data delay with sub-ns resolution in registers 3.1811 and 3.1812</u><br><u>0 = PCS does not provide information on receive path data delay with sub-ns resolution</u>   | <u>RO</u>        |
| 3.1800.1                                    | TimeSync transmit path data delay <u>ability, in ns</u>     | 1 = PCS provides information on transmit path data delay <u>with ns resolution</u> in registers 3.1801 through 3.1804<br>0 = PCS does not provide information on transmit path data delay <u>with ns resolution</u>     | RO               |
| 3.1800.0                                    | TimeSync receive path data delay <u>ability, in ns</u>      | 1 = PCS provides information on receive path data delay <u>with ns resolution</u> in registers 3.1805 through 3.1808<br>0 = PCS does not provide information on receive path data delay <u>with ns resolution</u>       | RO               |

<sup>a</sup>RO = Read only

*Insert new subclauses 45.2.3.67.1 through 45.2.3.67.7 at the end of 45.2.3.67 as follows:*

#### **45.2.3.67.1 Data delay measurement point ability (3.1800.13:12)**

Bits 3.1800.13 and 3.1800.12 indicate the PCS support of the beginning of the SFD, the beginning of the first symbol after the SFD, or both, as the data delay measurement point (see 90.7).

When bit 3.1800.12 is read as zero, the PCS supports only the beginning of the SFD.

When bit 3.1800.12 is read as one and bit 3.1800.13 is read as zero, the PCS supports only the beginning of the first symbol after the SFD.

When both bit 3.1800.12 and bit 3.1800.13 are read as one, the PCS supports both the beginning of the SFD and the beginning of the first symbol after the SFD. In this case, the data delay measurement point is selected by bit 3.1813.13 (see 45.2.3.69a.1).

#### **45.2.3.67.2 Multilane ability (3.1800.11)**

When read as a one, bit 3.1800.11 indicates that the PCS is able to report transmit and receive path data delays for multiple PCS lanes using the method recommended in 90.7.1 and 90A.4.

When read as a zero, bit 3.1800.11 indicates that the PCS is not able to report transmit and receive path data delays for multiple PCS lanes using the method recommended in 90.7.1 and 90A.4.

#### **45.2.3.67.3 PCS dynamic path data delay ability (3.1800.10)**

When read as a one, bit 3.1800.10 indicates that the PCS is able to report PCS dynamic path data delay (PDPDD) as TX\_NUM\_BIT\_CHANGE and RX\_NUM\_BIT\_CHANGE values (see 90.5.3 and 90.5.4) to the gRS.

When read as a zero, bit 3.1800.10 indicates that the PCS is not able to report PCS dynamic path data delay (PDPDD) as TX\_NUM\_BIT\_CHANGE and RX\_NUM\_BIT\_CHANGE values (see 90.5.3 and 90.5.4) to the gRS.

#### **45.2.3.67.4 TimeSync transmit path data delay ability, in sub-ns (3.1800.3)**

When read as a one, bit 3.1800.3 indicates that the PCS supports the PCS transmit path data delay registers, with sub-ns resolution (3.1809 and 3.1810).

When read as a zero, bit 3.1800.3 indicates that the PCS does not support the PCS transmit path data delay registers, with sub-ns resolution.

#### **45.2.3.67.5 TimeSync receive path data delay ability, in sub-ns (3.1800.2)**

When read as a one, bit 3.1800.2 indicates that the PCS supports the PCS receive path data delay registers, with sub-ns resolution (3.1811 and 3.1812).

When read as a zero, bit 3.1800.2 indicates that the PCS does not support the PCS receive path data delay registers, with sub-ns resolution.

#### **45.2.3.67.6 TimeSync transmit path data delay ability, in ns (3.1800.1)**

When read as a one, bit 3.1800.1 indicates that the PCS supports the PCS transmit path data delay registers, with ns resolution (3.1801 through 3.1804).

When read as a zero, bit 3.1800.1 indicates that the PCS does not support the PCS transmit path data delay registers, with ns resolution (3.1801 through 3.1804).

#### **45.2.3.67.7 TimeSync receive path data delay ability, in ns (3.1800.0)**

When read as a one, bit 3.1800.0 indicates that the PCS supports the PCS receive path data delay registers (3.1805 through 3.1808).

When read as a zero, bit 3.1800.0 indicates that the PCS does not support the PCS receive path data delay registers (3.1805 through 3.1808).

*Change the title and text of 45.2.3.68 as follows:*

#### **45.2.3.68 TimeSync PCS transmit path data delay (Registers 3.1801, 3.1802, 3.1803, 3.1804, 3.1809, and 3.1810)**

~~The TimeSync PCS transmit path data delay register contains the maximum (registers 3.1801, 3.1802, see Table 45-294) and minimum (registers 3.1803, 3.1804, see Table 45-294) values of the transmit path data delay. The transmit path data delay is expressed in units of ns. The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 1.1 (see 45.2.1.2.4).~~

The TimeSync PCS transmit path data delay registers contain the maximum and minimum values of the PCS transmit path data delay (see 90.7). The values contained in these registers are valid when the link is

established, as indicated by bit 2 in register 3.1 (see 45.2.3.2.7), and when the corresponding capability register bit (see Table 45–293) is asserted.

The maximum PCS transmit path data delay value is provided in two components. The first component (registers 3.1801 and 3.1802, see Table 45–294) provides the integer nanoseconds portion of the maximum PCS transmit path data delay, in units of nanoseconds. The second component (register 3.1809, see Table 45–294) provides the fractional nanoseconds portion of the maximum PCS transmit path data delay, in units of  $2^{-16}$  ns. If both components are valid (registers 3.1800.1 and 3.1800.3, see Table 45–293), the maximum PCS transmit path data delay is the sum of the values from these two components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the maximum PCS transmit path data delay.

The minimum PCS transmit path data delay value is provided in two components. The first component (registers 3.1803 and 3.1804, see Table 45–294) provides the integer nanoseconds portion of the minimum PCS transmit path data delay, in units of nanoseconds. The second component (register 3.1810, see Table 45–294) provides the fractional nanoseconds portion of the minimum PCS transmit path data delay, in units of  $2^{-16}$  ns. If both components are valid (registers 3.1800.1 and 3.1800.3, see Table 45–293), the minimum PCS transmit path data delay is the sum of the values from these two components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the minimum PCS transmit path data delay.

*Change Table 45–294 as follows:*

**Table 45–294—TimeSync PCS transmit path data delay register**

| Bit(s)             | Name                                                      | Description                                                               | R/W <sup>a</sup> |
|--------------------|-----------------------------------------------------------|---------------------------------------------------------------------------|------------------|
| 3.1801.15:0        | Maximum PCS transmit path data delay <u>in ns</u> , lower | <del>PCS_delay_TX_max [15:0]</del><br><u>PCS_delay_ns_TX_max [15:0]</u>   | RO, MW           |
| 3.1802.15:0        | Maximum PCS transmit path data delay <u>in ns</u> , upper | <del>PCS_delay_TX_max [31:16]</del><br><u>PCS_delay_ns_TX_max [31:16]</u> | RO, MW           |
| 3.1803.15:0        | Minimum PCS transmit path data delay <u>in ns</u> , lower | <del>PCS_delay_TX_min [15:0]</del><br><u>PCS_delay_ns_TX_min [15:0]</u>   | RO, MW           |
| 3.1804.15:0        | Minimum PCS transmit path data delay <u>in ns</u> , upper | <del>PCS_delay_TX_min [31:16]</del><br><u>PCS_delay_ns_TX_min [31:16]</u> | RO, MW           |
| <u>3.1809.15:0</u> | <u>Maximum PCS transmit path data delay in sub-ns</u>     | <u>PCS_delay_subns_TX_max [15:0]</u>                                      | <u>RO</u>        |
| <u>3.1810.15:0</u> | <u>Minimum PCS transmit path data delay in sub-ns</u>     | <u>PCS_delay_subns_TX_min [15:0]</u>                                      | <u>RO</u>        |

<sup>a</sup>RO = Read only, MW = Multi-word

*Change the title and text of 45.2.3.69 as follows:*

#### **45.2.3.69 TimeSync PCS receive path data delay (Registers 3.1805, 3.1806, 3.1807, 3.1808, 3.1811, and 3.1812)**

The TimeSync PCS receive path data delay register contains the maximum (registers 3.1805, 3.1806, see Table 45–295) and minimum (registers 3.1807, 3.1808, see Table 45–295) values of the receive path data delay. The receive path data delay is expressed in units of ns. The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 1.1 (see 45.2.1.2.4).

The TimeSync PCS receive path data delay registers contain the maximum and minimum values of the PCS receive path data delay (see 90.7). The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 3.1 (see 45.2.3.2.7), and when the corresponding capability register bit (see Table 45–293) is asserted.

The maximum PCS receive path data delay value is provided in two components. The first component (registers 3.1805 and 3.1806, see Table 45–295) provides the integer nanoseconds portion of the maximum PCS receive path data delay, in units of nanoseconds. The second component (register 3.1811, see Table 45–295) provides the fractional nanoseconds portion of the maximum PCS receive path data delay, in units of  $2^{-16}$  ns. If both components are valid (registers 3.1800.0 and 3.1800.2, see Table 45–293), the maximum PCS receive path data delay is the sum of the values from these two components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the maximum PCS receive path data delay.

The minimum PCS receive path data delay value is provided in two components. The first component (registers 3.1807 and 3.1808, see Table 45–295) provides the integer nanoseconds portion of the minimum PCS receive path data delay, in units of nanoseconds. The second component (register 3.1812, see Table 45–295) provides the fractional nanoseconds portion of the minimum PCS receive path data delay, in units of  $2^{-16}$  ns. If both components are valid (registers 3.1800.0 and 3.1800.2, see Table 45–293), the minimum PCS receive path data delay is the sum of the values from these two components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the minimum PCS receive path data delay.

*Change Table 45–295 as follows:*

**Table 45–295—TimeSync PCS receive path data delay register**

| Bit(s)             | Name                                                     | Description                                                           | R/W <sup>a</sup> |
|--------------------|----------------------------------------------------------|-----------------------------------------------------------------------|------------------|
| 3.1805.15:0        | Maximum PCS receive path data delay <u>in ns</u> , lower | <u>PCS_delay_RX_max [15:0]</u><br><u>PCS_delay_ns_RX_max [15:0]</u>   | RO, MW           |
| 3.1806.15:0        | Maximum PCS receive path data delay <u>in ns</u> , upper | <u>PCS_delay_RX_max [31:16]</u><br><u>PCS_delay_ns_RX_max [31:16]</u> | RO, MW           |
| 3.1807.15:0        | Minimum PCS receive path data delay <u>in ns</u> , lower | <u>PCS_delay_RX_min [15:0]</u><br><u>PCS_delay_ns_RX_min [15:0]</u>   | RO, MW           |
| 3.1808.15:0        | Minimum PCS receive path data delay <u>in ns</u> , upper | <u>PCS_delay_RX_min [31:16]</u><br><u>PCS_delay_ns_RX_min [31:16]</u> | RO, MW           |
| <u>3.1811.15:0</u> | <u>Maximum PCS receive path data delay in sub-ns</u>     | <u>PCS_delay_subns_RX_max [15:0]</u>                                  | <u>RO</u>        |
| <u>3.1812.15:0</u> | <u>Minimum PCS receive path data delay in sub-ns</u>     | <u>PCS_delay_subns_RX_min [15:0]</u>                                  | <u>RO</u>        |

<sup>a</sup>RO = Read only, MW = Multi-word

*Insert 45.2.3.69a (and its subclause) after 45.2.3.69 as follows:*

#### **45.2.3.69a TimeSync PCS configuration (Register 3.1813)**

This register is used to configure the data delay information provided by the PCS in support of a TimeSync Client. The assignment of bits in the TimeSync PCS configuration register is shown in Table 45–295a.

**Table 45–295a—TimeSync PCS configuration**

| Bit(s)       | Name                         | Description                                                                                                                                                                                                     | R/W <sup>a</sup> |
|--------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 3.1813.15:14 | Reserved                     | Value always 0                                                                                                                                                                                                  | RO               |
| 3.1813.13    | Data delay measurement point | 0 = Configures the PCS to use the data delay measurement point at the beginning of the SFD<br>1 = Configures the PCS to use the data delay measurement point at the beginning of the first symbol after the SFD | R/W              |
| 3.1813.12:0  | Reserved                     | Value always 0                                                                                                                                                                                                  | RO               |

<sup>a</sup>RO = Read only, R/W = Read/Write

#### 45.2.3.69a.1 Data delay measurement point (3.1813.13)

Bit 3.1813.13 is used to select the data delay measurement point (see 90.7).

When this bit is set to zero, the beginning of the SFD is used as the data delay measurement point.

When this bit is set to one, the beginning of the first symbol after the SFD is used as the data delay measurement point.

This bit has an effect only if the data delay measurement point ability bits are read as 11 in the TimeSync PCS capability register (see 45.2.3.67).

#### 45.2.4 PHY XS registers

*Change the rows for register addresses 4.1801 to 4.32767 in Table 45–314 as follows (unchanged rows not shown):*

**Table 45–314—PHY XS registers**

| Register address         | Register name                                             | Subclause        |
|--------------------------|-----------------------------------------------------------|------------------|
| ...                      |                                                           |                  |
| 4.1801 through 4.1804    | TimeSync PHY XS transmit path data delay <u>in ns</u>     | 45.2.4.29        |
| 4.1805 through 4.1808    | TimeSync PHY XS receive path data delay <u>in ns</u>      | 45.2.4.30        |
| <u>4.1809 and 4.1810</u> | <u>TimeSync PHY XS transmit path data delay in sub-ns</u> | <u>45.2.4.29</u> |
| <u>4.1811 and 4.1812</u> | <u>TimeSync PHY XS receive path data delay in sub-ns</u>  | <u>45.2.4.30</u> |
| 4.1813 through 4.32767   | Reserved                                                  |                  |
| ...                      |                                                           |                  |

#### 45.2.4.28 TimeSync PHY XS capability (Register 4.1800)

*Change the text of 45.2.4.28 as follows:*

The TimeSync PHY XS capability register (see Table 45–336) indicates the capability of the PHY XS to report the transmit and receive path data delay, stored in registers 4.1801 through 4.1804 and 4.1805 through 4.1808, respectively (with ns resolution in registers 4.1801 through 4.1804 and, separately, with sub-ns resolution in registers 4.1809 and 4.1810) and receive path data delay (with ns resolution in registers 4.1805 through 4.1808 and, separately, with sub-ns resolution in registers 4.1811 and 4.1812). See 90.7.

*Change Table 45–336 as follows:*

**Table 45–336—TimeSync PHY XS capability**

| Bit(s)                  | Name                                                        | Description                                                                                                                                                                                                                   | R/W <sup>a</sup> |
|-------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 4.1800.15:4<br><u>2</u> | Reserved                                                    | Value always 0                                                                                                                                                                                                                | RO               |
| <u>4.1800.3</u>         | <u>TimeSync transmit path data delay ability, in sub-ns</u> | <u>1 = PHY XS provides information on transmit path data delay with sub-ns resolution in registers 4.1809 and 4.1810</u><br><u>0 = PHY XS does not provide information on transmit path data delay with sub-ns resolution</u> | <u>RO</u>        |
| <u>4.1800.2</u>         | <u>TimeSync receive path data delay ability, in sub-ns</u>  | <u>1 = PHY XS provides information on receive path data delay with sub-ns resolution in registers 4.1811 and 4.1812</u><br><u>0 = PHY XS does not provide information on receive path data delay with sub-ns resolution</u>   | <u>RO</u>        |
| 4.1800.1                | TimeSync transmit path data delay <u>ability, in ns</u>     | 1 = PHY XS provides information on transmit path data delay <u>with ns resolution</u> in registers 4.1801 through 4.1804<br>0 = PHY XS does not provide information on transmit path data delay <u>with ns resolution</u>     | RO               |
| 4.1800.0                | TimeSync receive path data delay <u>ability, in ns</u>      | 1 = PHY XS provides information on receive path data delay <u>with ns resolution</u> in registers 4.1805 through 4.1808<br>0 = PHY XS does not provide information on receive path data delay <u>with ns resolution</u>       | RO               |

<sup>a</sup>RO = Read only

*Change the title and text of 45.2.4.29 as follows:*

**45.2.4.29 TimeSync PHY XS transmit path data delay (Registers 4.1801, 4.1802, 4.1803, 4.1804, 4.1809, and 4.1810)**

The TimeSync PHY XS transmit path data delay register contains the maximum (registers 4.1801, 4.1802, see Table 45–337) and minimum (registers 4.1803, 4.1804, see Table 45–337) values of the transmit path data delay. The transmit path data delay is expressed in units of ns. The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 1.1 (see 45.2.1.2.4).

The TimeSync PHY XS transmit path data delay registers contain the maximum and minimum values of the PHY XS transmit path data delay (see 90.7). The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 4.1 (see 45.2.4.2.7), and when the corresponding capability register bit (see Table 45–336) is asserted.

The maximum PHY XS transmit path data delay value is provided in two components. The first component (registers 4.1801 and 4.1802, see Table 45–337) provides the integer nanoseconds portion of the maximum PHY XS transmit path data delay, in units of nanoseconds. The second component (register 4.1809, see Table 45–337) provides the fractional nanoseconds portion of the maximum PHY XS transmit path data delay, in units of  $2^{-16}$  ns. If both components are valid (registers 4.1800.1 and 4.1800.3, see Table 45–336), the maximum PHY XS transmit path data delay is the sum of the values from these two components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the maximum PHY XS transmit path data delay.

The minimum PHY XS transmit path data delay value is provided in two components. The first component (registers 4.1803 and 4.1804, see Table 45–337) provides the integer nanoseconds portion of the minimum PHY XS transmit path data delay, in units of nanoseconds. The second component (register 4.1810, see Table 45–337) provides the fractional nanoseconds portion of the minimum PHY XS transmit path data

delay, in units of  $2^{-16}$  ns. If both components are valid (registers 4.1800.1 and 4.1800.3, see Table 45–336), the minimum PHY XS transmit path data delay is the sum of the values from these two components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the minimum PHY XS transmit path data delay.

*Change Table 45–337 as follows:*

**Table 45–337—TimeSync PHY XS transmit path data delay register**

| Bit(s)             | Name                                                     | Description                                                                         | R/W <sup>a</sup> |
|--------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------|------------------|
| 4.1801.15:0        | Maximum PHY XS transmit path data delay in ns, lower     | <del>PHY_XS_delay_TX_max [15:0]</del><br><del>PHY_XS_delay_ns_TX_max [15:0]</del>   | RO, MW           |
| 4.1802.15:0        | Maximum PHY XS transmit path data delay in ns, upper     | <del>PHY_XS_delay_TX_max [31:16]</del><br><del>PHY_XS_delay_ns_TX_max [31:16]</del> | RO, MW           |
| 4.1803.15:0        | Minimum PHY XS transmit path data delay in ns, lower     | <del>PHY_XS_delay_TX_min [15:0]</del><br><del>PHY_XS_delay_ns_TX_min [15:0]</del>   | RO, MW           |
| 4.1804.15:0        | Minimum PHY XS transmit path data delay in ns, upper     | <del>PHY_XS_delay_TX_min [31:16]</del><br><del>PHY_XS_delay_ns_TX_min [31:16]</del> | RO, MW           |
| <u>4.1809.15:0</u> | <u>Maximum PHY XS transmit path data delay in sub-ns</u> | <u>PHY_XS_delay_subns_TX_max [15:0]</u>                                             | <u>RO</u>        |
| <u>4.1810.15:0</u> | <u>Minimum PHY XS transmit path data delay in sub-ns</u> | <u>PHY_XS_delay_subns_TX_min [15:0]</u>                                             | <u>RO</u>        |

<sup>a</sup>RO = Read only, MW = Multi-word

*Change the title and text of 45.2.4.30 as follows:*

#### **45.2.4.30 TimeSync PHY XS receive path data delay (Registers 4.1805, 4.1806, 4.1807, 4.1808, 4.1811, and 4.1812)**

The TimeSync PHY XS receive path data delay register contains the maximum (registers 4.1805, 4.1806, see Table 45–338) and minimum (registers 4.1807, 4.1808, see Table 45–338) values of the receive path data delay. The receive path data delay is expressed in units of ns. The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 4.1 (see 45.2.4.2.7).

The TimeSync PHY XS receive path data delay registers contain the maximum and minimum values of the PHY XS receive path data delay (see 90.7). The values contained in these registers are valid when the PHY XS is not in reset or powered down as indicated by bits 15 and 11 in register 4.0 (see 45.2.4.1), and when the corresponding capability register bit (see Table 45–336) is asserted.

The maximum PHY XS receive path data delay value is provided in two components. The first component (registers 4.1805 and 4.1806, see Table 45–338) provides the integer nanoseconds portion of the maximum PHY XS receive path data delay, in units of nanoseconds. The second component (register 4.1811, see Table 45–338) provides the fractional nanoseconds portion of the maximum PHY XS receive path data delay, in units of  $2^{-16}$  ns. If both components are valid (registers 4.1800.0 and 4.1800.2, see Table 45–336), the maximum PHY XS receive path data delay is the sum of the values from these two components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the maximum PHY XS receive path data delay.

The minimum PHY XS receive path data delay value is provided in two components. The first component (registers 4.1807 and 4.1808, see Table 45–338) provides the integer nanoseconds portion of the minimum PHY XS receive path data delay, in units of nanoseconds. The second component (register 4.1812, see

Table 45–338) provides the fractional nanoseconds portion of the minimum PHY XS receive path data delay, in units of  $2^{-16}$  ns. If both components are valid (registers 4.1800.0 and 4.1800.2, see Table 45–336), the minimum PHY XS receive path data delay is the sum of the values from these two components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the minimum PHY XS receive path data delay.

*Change Table 45–338 as follows:*

**Table 45–338—TimeSync PHY XS receive path data delay register**

| Bit(s)             | Name                                                    | Description                                                                     | R/W <sup>a</sup> |
|--------------------|---------------------------------------------------------|---------------------------------------------------------------------------------|------------------|
| 4.1805.15:0        | Maximum PHY XS receive path data delay, in ns, lower    | <del>PHY_XS_delay_RX_max [15:0]</del><br><u>PHY_XS_delay_ns_RX_max [15:0]</u>   | RO, MW           |
| 4.1806.15:0        | Maximum PHY XS receive path data delay, in ns, upper    | <del>PHY_XS_delay_RX_max [31:16]</del><br><u>PHY_XS_delay_ns_RX_max [31:16]</u> | RO, MW           |
| 4.1807.15:0        | Minimum PHY XS receive path data delay, in ns, lower    | <del>PHY_XS_delay_RX_min [15:0]</del><br><u>PHY_XS_delay_ns_RX_min [15:0]</u>   | RO, MW           |
| 4.1808.15:0        | Minimum PHY XS receive path data delay, in ns, upper    | <del>PHY_XS_delay_RX_min [31:16]</del><br><u>PHY_XS_delay_ns_RX_min [31:16]</u> | RO, MW           |
| <u>4.1811.15:0</u> | <u>Maximum PHY XS receive path data delay in sub-ns</u> | <u>PHY_XS_delay_subns_RX_max [15:0]</u>                                         | <u>RO</u>        |
| <u>4.1812.15:0</u> | <u>Minimum PHY XS receive path data delay in sub-ns</u> | <u>PHY_XS_delay_subns_RX_min [15:0]</u>                                         | <u>RO</u>        |

<sup>a</sup>RO = Read only, MW = Multi-word

#### 45.2.5 DTE XS registers

*Change the rows for register addresses 5.1801 to 5.32767 in Table 45–339 as follows (unchanged rows not shown):*

**Table 45–339—DTE XS registers**

| Register address                     | Register name                                             | Subclause        |
|--------------------------------------|-----------------------------------------------------------|------------------|
| ...                                  |                                                           |                  |
| 5.1801 through 5.1804                | TimeSync DTE XS transmit path data delay <u>in ns</u>     | 45.2.5.29        |
| 5.1805 through 5.1808                | TimeSync DTE XS receive path data delay <u>in ns</u>      | 45.2.5.30        |
| <u>5.1809 and 5.1810</u>             | <u>TimeSync DTE XS transmit path data delay in sub-ns</u> | <u>45.2.5.29</u> |
| <u>5.1811 and 5.1812</u>             | <u>TimeSync DTE XS receive path data delay in sub-ns</u>  | <u>45.2.5.30</u> |
| <u>5.1813</u>                        | <u>TimeSync DTE XS configuration</u>                      | <u>45.2.5.31</u> |
| 5.1814 <del>09</del> through 5.32767 | Reserved                                                  |                  |
| ...                                  |                                                           |                  |

#### 45.2.5.28 TimeSync DTE XS capability (Register 5.1800)

*Change the text of 45.2.5.28 as follows:*

~~The TimeSync DTE XS capability register (see Table 45–361) indicates. This register is used to indicate the capability of the DTE XS to report provide the transmit and receive path data delay information in support of a TimeSync Client. See 90.7, stored in registers 5.1801 through 5.1804 and 5.1805 through 5.1808, respectively. The assignment of bits in the TimeSync DTE XS capability register is shown in Table 45–361.~~

*Change Table 45–361 as follows:*

**Table 45–361—TimeSync DTE XS capability**

| Bit(s)            | Name                                                 | Description                                                                                                                                                                                                                                                                                                                                                         | R/W <sup>a</sup> |
|-------------------|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 5.1800.15:14<br>2 | Reserved                                             | Value always 0                                                                                                                                                                                                                                                                                                                                                      | RO               |
| 5.1800.13:12      | Data delay measurement point ability                 | <p>0 = DTE XS supports only the beginning of the SFD as the data delay measurement point</p> <p>01 = DTE XS supports only the beginning of the first symbol after the SFD as the data delay measurement point</p> <p>11 = DTE XS supports both the beginning of the SFD and the beginning of the first symbol after the SFD as the data delay measurement point</p> | RO               |
| 5.1800.11:4       | Reserved                                             | Value always 0                                                                                                                                                                                                                                                                                                                                                      | RO               |
| 5.1800.3          | TimeSync transmit path data delay ability, in sub-ns | <p>1 = DTE XS provides information on transmit path data delay with sub-ns resolution in registers 5.1809 and 5.1810</p> <p>0 = DTE XS does not provide information on transmit path data delay with sub-ns resolution</p>                                                                                                                                          | RO               |
| 5.1800.2          | TimeSync receive path data delay ability, in sub-ns  | <p>1 = DTE XS provides information on receive path data delay with sub-ns resolution in registers 5.1811 and 5.1812</p> <p>0 = DTE XS does not provide information on receive path data delay with sub-ns resolution</p>                                                                                                                                            | RO               |
| 5.1800.1          | TimeSync transmit path data delay ability, in ns     | <p>1 = DTE XS provides information on transmit path data delay with ns resolution in registers 5.1801 through 5.1804</p> <p>0 = DTE XS does not provide information on transmit path data delay with ns resolution</p>                                                                                                                                              | RO               |
| 5.1800.0          | TimeSync receive path data delay ability, in ns      | <p>1 = DTE XS provides information on receive path data delay with ns resolution in registers 5.1805 through 5.1808</p> <p>0 = DTE XS does not provide information on receive path data delay with ns resolution</p>                                                                                                                                                | RO               |

<sup>a</sup>RO = Read only

*Insert new subclauses 45.2.5.28.1 through 45.2.5.28.5 at the end of 45.2.5.28 as follows:*

##### 45.2.5.28.1 Data delay measurement point ability (5.1800.13:12)

Bits 5.1800.13 and 5.1800.12 indicate the DTE XS support of the beginning of the SFD, the beginning of the first symbol after the SFD, or both, as the data delay measurement point (see 90.7).

When bit 5.1800.12 is read as zero, the DTE XS supports only the beginning of the SFD.

When bit 5.1800.12 is read as one and bit 5.1800.13 is read as zero, the DTE XS supports only the beginning of the first symbol after the SFD.

When both bit 5.1800.12 and bit 5.1800.13 are read as one, the DTE XS supports both the beginning of the SFD and the beginning of the first symbol after the SFD. In this case, the data delay measurement point is selected by bit 5.1813.13 (see 45.2.5.31.1).

#### **45.2.5.28.2 TimeSync transmit path data delay ability, in sub-ns (5.1800.3)**

When read as a one, bit 5.1800.3 indicates that the DTE XS supports the DTE XS transmit path data delay registers, with sub-ns resolution (5.1809 and 5.1810).

When read as a zero, bit 5.1800.3 indicates that the DTE XS does not support the DTE XS transmit path data delay registers, with sub-ns resolution.

#### **45.2.5.28.3 TimeSync receive path data delay ability, in sub-ns (5.1800.2)**

When read as a one, bit 5.1800.2 indicates that the DTE XS supports the DTE XS receive path data delay registers, with sub-ns resolution (5.1811 and 5.1812).

When read as a zero, bit 5.1800.2 indicates that the DTE XS does not support the DTE XS receive path data delay registers, with sub-ns resolution.

#### **45.2.5.28.4 TimeSync transmit path data delay ability, in ns (5.1800.1)**

When read as a one, bit 5.1800.1 indicates that the DTE XS supports the DTE XS transmit path data delay registers, with ns resolution (5.1801 through 5.1804).

When read as a zero, bit 5.1800.1 indicates that the DTE XS does not support the DTE XS transmit path data delay registers, with ns resolution (5.1801 through 5.1804).

#### **45.2.5.28.5 TimeSync receive path data delay ability, in ns (5.1800.0)**

When read as a one, bit 5.1800.0 indicates that the DTE XS supports DTE XS receive path data delay registers, with ns resolution (5.1805 through 5.1808).

When read as a zero, bit 5.1800.0 indicates that the DTE XS does not support the DTE XS receive path data delay registers, with ns resolution (5.1805 through 5.1808).

*Change the title and text of 45.2.5.29 as follows:*

#### **45.2.5.29 TimeSync DTE XS transmit path data delay (Registers 5.1801, 5.1802, 5.1803, 5.1804, 5.1809, and 5.1810)**

~~The TimeSync DTE XS transmit path data delay register contains the maximum (registers 5.1801, 5.1802, see Table 45–362) and minimum (registers 5.1803, 5.1804, see Table 45–362) values of the transmit path data delay. The transmit path data delay is expressed in units of ns. The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 1.1 (see 45.2.1.2.4).~~

The TimeSync DTE XS transmit path data delay registers contain the maximum and minimum values of the DTE XS transmit path data delay (see 90.7). The values contained in these registers are valid when the DTE XS is not in reset or powered down as indicated by bits 15 and 11 in register 5.0 (see 45.2.5.1), and when the corresponding capability register bit (see Table 45–361) is asserted.

The maximum DTE XS transmit path data delay value is provided in two components. The first component (registers 5.1801 and 5.1802, see Table 45–362) provides the integer nanoseconds portion of the maximum DTE XS transmit path data delay, in units of nanoseconds. The second component (register 5.1809, see Table 45–362) provides the fractional nanoseconds portion of the maximum DTE XS transmit path data delay, in units of  $2^{-16}$  ns. If both components are valid (registers 5.1800.1 and 5.1800.3, see Table 45–361), the maximum DTE XS transmit path data delay is the sum of the values from these two components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the maximum DTE XS transmit path data delay.

The minimum DTE XS transmit path data delay value is provided in two components. The first component (registers 5.1803 and 5.1804, see Table 45–362) provides the integer nanoseconds portion of the minimum DTE XS transmit path data delay, in units of nanoseconds. The second component (register 5.1810, see Table 45–362) provides the fractional nanoseconds portion of the minimum DTE XS transmit path data delay, in units of  $2^{-16}$  ns. If both components are valid (registers 5.1800.1 and 5.1800.3, see Table 45–361), the minimum DTE XS transmit path data delay is the sum of the values from these two components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the minimum DTE XS transmit path data delay.

*Change Table 45–362 as follows:*

**Table 45–362—TimeSync DTE XS transmit path data delay register**

| Bit(s)             | Name                                                     | Description                                                                     | R/W <sup>a</sup> |
|--------------------|----------------------------------------------------------|---------------------------------------------------------------------------------|------------------|
| 5.1801.15:0        | Maximum DTE XS transmit path data delay, in ns, lower    | <del>DTE_XS_delay_TX_max [15:0]</del><br><u>DTE_XS_delay_ns_TX_max [15:0]</u>   | RO, MW           |
| 5.1802.15:0        | Maximum DTE XS transmit path data delay, in ns, upper    | <del>DTE_XS_delay_TX_max [31:16]</del><br><u>DTE_XS_delay_ns_TX_max [31:16]</u> | RO, MW           |
| 5.1803.15:0        | Minimum DTE XS transmit path data delay, in ns, lower    | <del>DTE_XS_delay_TX_min [15:0]</del><br><u>DTE_XS_delay_ns_TX_min [15:0]</u>   | RO, MW           |
| 5.1804.15:0        | Minimum DTE XS transmit path data delay, in ns, upper    | <del>DTE_XS_delay_TX_min [31:16]</del><br><u>DTE_XS_delay_ns_TX_min [31:16]</u> | RO, MW           |
| <u>5.1809.15:0</u> | <u>Maximum DTE XS transmit path data delay in sub-ns</u> | <u>DTE_XS_delay_subns_TX_max [15:0]</u>                                         | <u>RO</u>        |
| <u>5.1810.15:0</u> | <u>Minimum DTE XS transmit path data delay in sub-ns</u> | <u>DTE_XS_delay_subns_TX_min [15:0]</u>                                         | <u>RO</u>        |

<sup>a</sup>RO = Read only, MW = Multi-word

*Change the title and text of 45.2.5.30 as follows:*

#### **45.2.5.30 TimeSync DTE XS receive path data delay (Registers 5.1805, 5.1806, 5.1807, 5.1808, 5.1811, and 5.1812)**

The TimeSync DTE XS receive path data delay register contains the maximum (registers 5.1805, 5.1806, see Table 45–363) and minimum (registers 5.1807, 5.1808, see Table 45–363) values of the receive path data delay. The receive path data delay is expressed in units of ns. The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 5.1 (see 45.2.5.2.7).

The TimeSync DTE XS receive path data delay registers contain the maximum and minimum values of the DTE XS receive path data delay (see 90.7). The values contained in these registers are valid when the link is

established, as indicated by bit 2 in register 5.1 (see 45.2.5.2.7), and when the corresponding capability register bit (see Table 45–361) is asserted.

The maximum DTE XS receive path data delay value is provided in two components. The first component (registers 5.1805 and 5.1806, see Table 45–363) provides the integer nanoseconds portion of the maximum DTE XS receive path data delay, in units of nanoseconds. The second component (register 5.1811, see Table 45–363) provides the fractional nanoseconds portion of the maximum DTE XS receive path data delay, in units of  $2^{-16}$  ns. If both components are valid (registers 5.1800.0 and 5.1800.2, see Table 45–361), the maximum DTE XS receive path data delay is the sum of the values from these two components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the maximum DTE XS receive path data delay.

The minimum DTE XS receive path data delay value is provided in two components. The first component (registers 5.1807 and 5.1808, see Table 45–363) provides the integer nanoseconds portion of the minimum DTE XS receive path data delay, in units of nanoseconds. The second component (register 5.1812, see Table 45–363) provides the fractional nanoseconds portion of the minimum DTE XS receive path data delay, in units of  $2^{-16}$  ns. If both components are valid (registers 5.1800.0 and 5.1800.2, see Table 45–361), the minimum DTE XS receive path data delay is the sum of the values from these two components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the minimum DTE XS receive path data delay.

*Change Table 45–363 as follows:*

**Table 45–363—TimeSync DTE XS receive path data delay register**

| Bit(s)             | Name                                                    | Description                                                                         | R/W <sup>a</sup> |
|--------------------|---------------------------------------------------------|-------------------------------------------------------------------------------------|------------------|
| 5.1805.15:0        | Maximum DTE XS receive path data delay in ns, lower     | <del>DTE_XS_delay_RX_max</del> [15:0]<br><del>DTE_XS_delay_ns_RX_max</del> [15:0]   | RO, MW           |
| 5.1806.15:0        | Maximum DTE XS receive path data delay in ns, upper     | <del>DTE_XS_delay_RX_max</del> [31:16]<br><del>DTE_XS_delay_ns_RX_max</del> [31:16] | RO, MW           |
| 5.1807.15:0        | Minimum DTE XS receive path data delay in ns, lower     | <del>DTE_XS_delay_RX_min</del> [15:0]<br><del>DTE_XS_delay_ns_RX_min</del> [15:0]   | RO, MW           |
| 5.1808.15:0        | Minimum DTE XS receive path data delay in ns, upper     | <del>DTE_XS_delay_RX_min</del> [31:16]<br><del>DTE_XS_delay_ns_RX_min</del> [31:16] | RO, MW           |
| <u>5.1811.15:0</u> | <u>Maximum DTE XS receive path data delay in sub-ns</u> | <u>DTE_XS_delay_subns_RX_max</u> [15:0]                                             | <u>RO</u>        |
| <u>5.1812.15:0</u> | <u>Minimum DTE XS receive path data delay in sub-ns</u> | <u>DTE_XS_delay_subns_RX_min</u> [15:0]                                             | <u>RO</u>        |

<sup>a</sup>RO = Read only, MW = Multi-word

*Insert 45.2.5.31 (and its subclause) after 45.2.5.30 as follows:*

#### **45.2.5.31 TimeSync DTE XS configuration (Register 5.1813)**

This register is used to configure the path data delay information provided by the DTE XS in support of a TimeSync Client. The assignment of bits in the TimeSync DTE XS configuration register is shown in Table 45–363a.

**Table 45–363a—TimeSync DTE XS configuration**

| Bit(s)       | Name                         | Description                                                                                                                                                                                                         | R/W <sup>a</sup> |
|--------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 5.1813.15:14 | Reserved                     | Value always 0                                                                                                                                                                                                      | RO               |
| 5.1813.13    | Data delay measurement point | 0 = DTE XS is configured to use the data delay measurement point at the beginning of the SFD<br>1 = DTE XS is configured to use the data delay measurement point at the beginning of the first symbol after the SFD | R/W              |
| 5.1813.12:0  | Reserved                     | Value always 0                                                                                                                                                                                                      | RO               |

<sup>a</sup>RO = Read only, R/W = Read/Write

#### 45.2.5.31.1 Data delay measurement point (5.1813.13)

Bit 5.1813.13 is used to select the data delay measurement point (see 90.7).

When this bit is set to zero, the beginning of the SFD is used as the data delay measurement point.

When this bit is set to one, the beginning of the first symbol after the SFD is used as the data delay measurement point.

This bit has an effect only if the data delay measurement point ability bits are read as 11 in the TimeSync DTE XS capability register (see 45.2.5.28).

#### 45.2.6 TC registers

*Change the rows for register addresses 6.1801 to 6.32767 in Table 45–364 as follows (unchanged rows not shown):*

**Table 45–364—TC registers**

| Register address                     | Register name                                         | Subclause        |
|--------------------------------------|-------------------------------------------------------|------------------|
| ...                                  |                                                       |                  |
| 6.1801 through 6.1804                | TimeSync TC transmit path data delay <u>in ns</u>     | 45.2.6.15        |
| 6.1805 through 6.1808                | TimeSync TC receive path data delay <u>in ns</u>      | 45.2.6.16        |
| <u>6.1809 and 6.1810</u>             | <u>TimeSync TC transmit path data delay in sub-ns</u> | <u>45.2.6.15</u> |
| <u>6.1811 and 6.1812</u>             | <u>TimeSync TC receive path data delay in sub-ns</u>  | <u>45.2.6.16</u> |
| 6.1813 <del>09</del> through 6.32767 | Reserved                                              |                  |
| ...                                  |                                                       |                  |

#### 45.2.6.14 TimeSync TC capability (Register 6.1800)

*Change the text of 45.2.6.14 as follows:*

The TimeSync TC capability register (see Table 45–375) indicates the capability of the TC to report the transmit ~~and receive path~~ data delay, ~~stored in registers 6.1801 through 6.1804 and 6.1805 through 6.1808, respectively~~ (with ns resolution in registers 6.1801 through 6.1804 and, separately, with sub-ns resolution in

registers 6.1809 and 6.1810) and receive path data delay (with ns resolution in registers 6.1805 through 6.1808 and, separately, with sub-ns resolution in registers 6.1811 and 6.1812). See 90.7.

*Change Table 45–375 as follows:*

**Table 45–375—TimeSync TC capability**

| Bit(s)           | Name                                                        | Description                                                                                                                                                                                                           | R/W <sup>a</sup> |
|------------------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 6.1800.15:4<br>2 | Reserved                                                    | Value always 0                                                                                                                                                                                                        | RO               |
| 6.1800.3         | <u>TimeSync transmit path data delay ability, in sub-ns</u> | <u>1 = TC provides information on transmit path data delay with sub-ns resolution in registers 6.1809 and 6.1810</u><br><u>0 = TC does not provide information on transmit path data delay with sub-ns resolution</u> | <u>RO</u>        |
| 6.1800.2         | <u>TimeSync receive path data delay ability, in sub-ns</u>  | <u>1 = TC provides information on receive path data delay with sub-ns resolution in registers 6.1811 and 6.1812</u><br><u>0 = TC does not provide information on receive path data delay with sub-ns resolution</u>   | <u>RO</u>        |
| 6.1800.1         | TimeSync transmit path data delay <u>ability, in ns</u>     | 1 = TC provides information on transmit path data delay <u>with ns resolution</u> in registers 6.1801 through 6.1804<br>0 = TC does not provide information on transmit path data delay <u>with ns resolution</u>     | RO               |
| 6.1800.0         | TimeSync receive path data delay <u>ability, in ns</u>      | 1 = TC provides information on receive path data delay <u>with ns resolution</u> in registers 6.1805 through 6.1808<br>0 = TC does not provide information on receive path data delay <u>with ns resolution</u>       | RO               |

<sup>a</sup>RO = Read only

*Change the title and text of 45.2.6.15 as follows:*

#### **45.2.6.15 TimeSync TC transmit path data delay (Registers 6.1801, 6.1802, 6.1803, 6.1804, 6.1809, and 6.1810)**

~~The TimeSync TC transmit path data delay register contains the maximum (registers 6.1801, 6.1802, see Table 45–376) and minimum (registers 6.1803, 6.1804, see Table 45–376) values of the transmit path data delay. The transmit path data delay is expressed in units of ns. The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 1.1 (see 45.2.1.2.4).~~

The TimeSync TC transmit path data delay registers contain the maximum and minimum values of the TC transmit path data delay (see 90.7). The values contained in these registers are valid when the TC is not in reset as indicated by bit 15 in register 6.0 (see 45.2.6.1.1), and when the corresponding capability register bit (see Table 45–375) is asserted.

The maximum TC transmit path data delay value is provided in two components. The first component (registers 6.1801 and 6.1802, see Table 45–376) provides the integer nanoseconds portion of the maximum TC transmit path data delay, in units of nanoseconds. The second component (register 6.1809, see Table 45–376) provides the fractional nanoseconds portion of the maximum TC transmit path data delay, in units of  $2^{-16}$  ns. If both components are valid (registers 6.1800.1 and 6.1800.3, see Table 45–375), the maximum TC transmit path data delay is the sum of the values from these two components. If either of the

two components is not valid, then the value corresponding to the invalid component is not included in the maximum TC transmit path data delay.

The minimum TC transmit path data delay value is provided in two components. The first component (registers 6.1803 and 6.1804, see Table 45–376) provides the integer nanoseconds portion of the minimum TC transmit path data delay, in units of nanoseconds. The second component (register 6.1810, see Table 45–376) provides the fractional nanoseconds portion of the minimum TC transmit path data delay, in units of  $2^{-16}$  ns. If both components are valid (registers 6.1800.1 and 6.1800.3, see Table 45–375), the minimum TC transmit path data delay is the sum of the values from these two components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the minimum TC transmit path data delay.

*Change Table 45–376 as follows:*

**Table 45–376—TimeSync TC transmit path data delay register**

| Bit(s)      | Name                                             | Description                                                         | R/W <sup>a</sup> |
|-------------|--------------------------------------------------|---------------------------------------------------------------------|------------------|
| 6.1801.15:0 | Maximum TC transmit path data delay in ns, lower | <u>TC_delay_TX_max [15:0]</u><br><u>TC_delay_ns_TX_max [15:0]</u>   | RO, MW           |
| 6.1802.15:0 | Maximum TC transmit path data delay in ns, upper | <u>TC_delay_TX_max [31:16]</u><br><u>TC_delay_ns_TX_max [31:16]</u> | RO, MW           |
| 6.1803.15:0 | Minimum TC transmit path data delay in ns, lower | <u>TC_delay_TX_min [15:0]</u><br><u>TC_delay_ns_TX_min [15:0]</u>   | RO, MW           |
| 6.1804.15:0 | Minimum TC transmit path data delay in ns, upper | <u>TC_delay_TX_min [31:16]</u><br><u>TC_delay_ns_TX_min [31:16]</u> | RO, MW           |
| 6.1809.15:0 | Maximum TC transmit path data delay in sub-ns    | <u>TC_delay_subns_TX_max [15:0]</u>                                 | RO               |
| 6.1810.15:0 | Minimum TC transmit path data delay in sub-ns    | <u>TC_delay_subns_TX_min [15:0]</u>                                 | RO               |

<sup>a</sup>RO = Read only, MW = Multi-word

*Change the title and text of 45.2.6.16 as follows:*

#### **45.2.6.16 TimeSync TC receive path data delay (Registers 6.1805, 6.1806, 6.1807, 6.1808, 6.1811, and 6.1812)**

~~The TimeSync TC receive path data delay register contains the maximum (registers 6.1805, 6.1806, see Table 45–377) and minimum (registers 6.1807, 6.1808, see Table 45–377) values of the receive path delay. The receive path data delay is expressed in units of ns. The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 1.1 (see 45.2.1.2.4).~~

The TimeSync TC receive path data delay registers contain the maximum and minimum values of the TC receive path data delay (see 90.7). The values contained in these registers are valid when the TC is not in reset as indicated by bit 15 in register 6.0 (see 45.2.6.1.1), and when the corresponding capability register bit (see Table 45–375) is asserted.

The maximum TC receive path data delay value is provided in two components. The first component (registers 6.1805 and 6.1806, see Table 45–377) provides the integer nanoseconds portion of the maximum TC receive path data delay, in units of nanoseconds. The second component (register 6.1811, see Table 45–377) provides the fractional nanoseconds portion of the maximum TC receive path data delay, in

units of  $2^{-16}$  ns. If both components are valid (registers 6.1800.0 and 6.1800.2, see Table 45–375), the maximum TC receive path data delay is the sum of the values from these two components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the maximum TC receive path data delay.

The minimum TC receive path data delay value is provided in two components. The first component (registers 6.1807 and 6.1808, see Table 45–377) provides the integer nanoseconds portion of the minimum TC receive path data delay, in units of nanoseconds. The second component (register 6.1812, see Table 45–377) provides the fractional nanoseconds portion of the minimum TC receive path data delay, in units of  $2^{-16}$  ns. If both components are valid (registers 6.1800.0 and 6.1800.2, see Table 45–375), the minimum TC receive path data delay is the sum of the values from these two components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the minimum TC receive path data delay.

*Change Table 45–377 as follows:*

**Table 45–377—TimeSync TC receive path data delay register**

| Bit(s)             | Name                                                    | Description                                                      | R/W <sup>a</sup> |
|--------------------|---------------------------------------------------------|------------------------------------------------------------------|------------------|
| 6.1805.15:0        | Maximum TC receive path data delay <u>in ns</u> , lower | <del>TC_delay_RX_max [15:0]</del><br>TC_delay_ns_RX_max [15:0]   | RO, MW           |
| 6.1806.15:0        | Maximum TC receive path data delay <u>in ns</u> , upper | <del>TC_delay_RX_max [31:16]</del><br>TC_delay_ns_RX_max [31:16] | RO, MW           |
| 6.1807.15:0        | Minimum TC receive path data delay <u>in ns</u> , lower | <del>TC_delay_RX_min [15:0]</del><br>TC_delay_ns_RX_min [15:0]   | RO, MW           |
| 6.1808.15:0        | Minimum TC receive path data delay <u>in ns</u> , upper | <del>TC_delay_RX_min [31:16]</del><br>TC_delay_ns_RX_min [31:16] | RO, MW           |
| <u>6.1811.15:0</u> | <u>Maximum TC receive path data delay in sub-ns</u>     | TC_delay_subns_RX_max [15:0]                                     | <u>RO</u>        |
| <u>6.1812.15:0</u> | <u>Minimum TC receive path data delay in sub-ns</u>     | TC_delay_subns_RX_min [15:0]                                     | <u>RO</u>        |

<sup>a</sup>RO = Read only, MW = Multi-word

## 90. Ethernet support for time synchronization protocols

### 90.2 Overview

*Change 90.2 as follows:*

The goal of this clause is to provide an accurate indication of the time at which any packet is transmitted or is received—transmission and reception initiation times of all packets, as required to support various time synchronization protocols, e.g., IEEE Std 1588-2008 [B41]; and IEEE Std 802.1AS [B39].

The specific goals are to:

- a) Define the Time Synchronization Service Interface (TSSI), with associated service primitives;
- b) Add management registers to indicate the maximum and minimum path data delays for estimation of link latency at the Time Synchronization Protocol (TimeSync) Client.

### 90.3 Relationship with other IEEE standards

*Change 90.3 as follows:*

~~Per 90.2, the TimeSync capability provides support for various time synchronization protocols, including e.g., IEEE Std 1588-2008 or IEEE Std 802.1AS. The definition of TimeSync Client, its capabilities and functions is outside the scope of IEEE Std 802.3.~~

The TSSI and path data delays registers provide support for time synchronization protocols, for example, IEEE Std 1588 [B41] and IEEE Std 802.1AS [B39]. The definition of the TimeSync Client, its capabilities, and its functions, is beyond the scope of this standard.

The path data delay in this standard is illustrated in Figure 90–3 and may be associated with the timestamping mechanisms in IEEE Std 1588 [B41] and IEEE Std 802.1AS [B39].

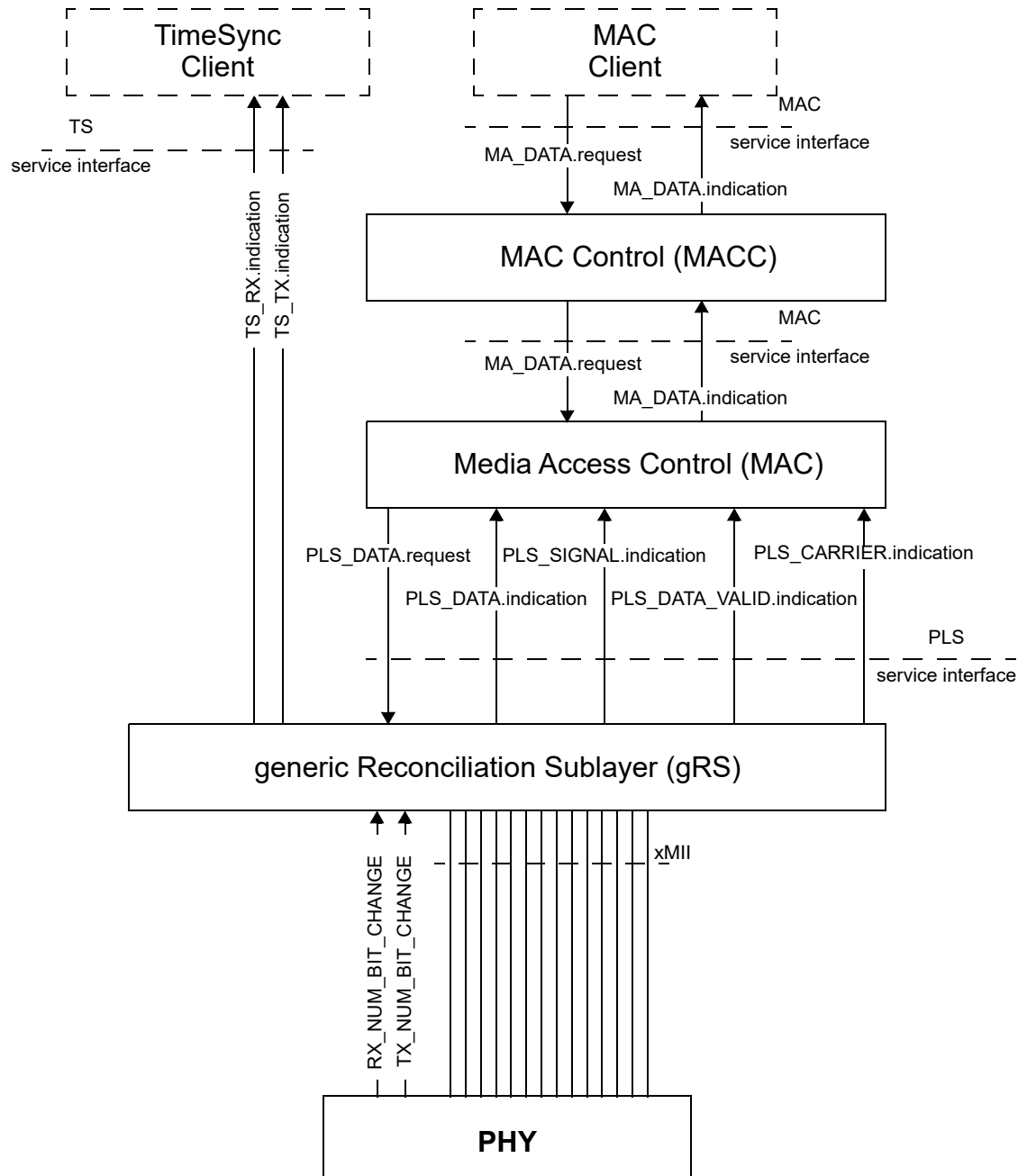
The path data delay of the PHY receiver and the PHY transmitter in this standard corresponds to the *ingressLatency* and the *egressLatency*, respectively, from IEEE Std 1588 [B41] and IEEE Std 802.1AS [B39]. The xMII of this standard corresponds to the timestamp measurement plane in IEEE Std 802.1AS [B39]. The MDI of this standard serves as the model for the timestamp generation reference plane in IEEE Std 1588 [B41] and IEEE Std 802.1AS [B39]. This is true for PHYs that have a constant path data delay and is modeled by the xMII timestamp measurement plane (see 90.5) and the reported transmit and receive path data delay values (see 90.7). However, for PHYs that contain functions with transmit intrinsic delay variations that are mirrored by corresponding receive intrinsic delay variations (see 90A.7), the timestamp generation reference plane is still modeled by the xMII timestamp measurement plane and the reported transmit and receive path data delay values because, as described in 90.7.1, 90A.4, and 90A.7, the correct results are produced for the end-to-end delay measurements.

## 90.4 Time Synchronization Service Interface (TSSI)

### 90.4.1 Introduction

#### 90.4.1.1 Interlayer service interfaces

Replace Figure 90–1 with the following figure:



**Figure 90–1—Relationship of the TimeSync Client, TSSI, and gRS sublayer relative to MAC and MAC Client and associated interfaces**

### 90.4.1.2 Responsibilities of TimeSync Client

*Change 90.4.1.2 as follows:*

The TimeSync Client ~~can~~ may use the indication of the ~~event corresponding to the egress and ingress of packets provided by the TSSI, combined with knowledge of the time synchronization protocol packets frames, to capture select the egress and ingress times-time of packets relevant to the protocol at the xMII. Which frames are~~ The identification of specific protocol packets of interest to any particular protocol is beyond the scope of this standard.

~~The TimeSync Client can use the indication of the event corresponding to the egress and ingress of packets at the xMII provided by the TSSI, together with the information provided by the TimeSync PHY transmit data delay and TimeSync PHY receive data delay (see 30.13.1).~~

When the TimeSync Client captures the egress time of a relevant packet at the xMII, it may be used along with the TimeSync PHY transmit path data delay information, if available, and the PCS dynamic transmit path data delay, if supplied, to calculate the egress time of the data delay measurement point (DDMP) at the MDI. The difference between the maximum and minimum PHY transmit path data delay values, if available, may be used by the TimeSync Client to determine the uncertainty of the calculated egress time at the MDI.

When the TimeSync Client captures the ingress time of a relevant packet at the xMII, it may be used along with the TimeSync PHY receive path data delay information, if available, and the PCS dynamic receive path data delay, if supplied, to calculate the ingress time of the DDMP at the MDI. The difference between the maximum and minimum PHY receive path data delay values, if available, may be used by the TimeSync Client to determine the uncertainty of the calculated ingress time at the MDI.

### 90.4.2 TSSI

*Change 90.4.2 as follows:*

The following specifies the service interface provided by the gRS to the TimeSync Client. ~~These~~ The services ~~provided by the primitives~~ are described in an abstract manner and do not imply any particular implementation. The model used in this service specification is identical to that used in 1.2.2.

The path data delay of a packet is measured using a specific location in the packet called the DDMP. The DDMP is either the beginning of the start-of-frame delimiter (SFD) or the beginning of the first symbol after the SFD in the packet, as selected by register bits 3.1813.13 and 5.1813.13. Within the scope of this clause, the term ‘first symbol after the SFD’ denotes the first octet after the SFD when referencing an xMII.

The following primitives are defined:

- TS\_TX.indication
- TS\_RX.indication

### 90.4.3 Detailed service specification

#### 90.4.3.1 TS\_TX.indication primitive

*Change 90.4.3.1 as follows:*

This primitive defines the transfer of the indication of an event between the gRS and the TimeSync Client. When asserted, the TimeSync Client is notified that a valid DDMP was detected by the gRS TS\_DDMP\_Detect\_TX function (see 90.5.1) in the xMII transmit signals.

### 90.4.3.1.1 Semantics

*Change 90.4.3.1.1 (as amended by IEEE Std 802.3de-2022) as follows:*

The semantics of the primitive are as follows:

TS\_TX.indication(~~SFD~~, DDMP, MM, PDPDD)

~~The SFD parameter can take only one possible value, DETECTED. When asserted (SFD = DETECTED), the TimeSync Client is notified that a valid SFD was detected by the gRS sublayer TS\_SFD\_Detect\_TX function (see 90.5.1) in the xMII transmit signals.~~

The DDMP parameter can take one of two possible values, SFD and FIRST\_SYMBOL. The value SFD indicates that the TS\_TX.indication primitive was generated as the result of the beginning of an SFD being transferred across the transmit path of the xMII. The value FIRST\_SYMBOL indicates that the TS\_TX.indication primitive was generated as the result of the beginning of the first symbol after an SFD being transferred across the transmit path of the xMII.

The information given by the DDMP parameter may be used by a TimeSync Client to get bit time resolution for the timestamp capture for the DDMP in the xMII word.

The use of the beginning of the SFD, or the beginning of the first symbol after the SFD, as the DDMP requires consistent configuration of both the gRS and the PHY (see 45.2.3.69a and 45.2.5.31) for correct operation.

~~The MM parameter is mandatory when the MAC Merge sublayer (see Clause 99) is instantiated. The MM parameter, when present, can take one of two possible values, i.e., PMAC or EMAC. The value EMAC indicates that the TS\_TX.indication primitive is associated with a packet that has an SMD-E (SFD) value has been detected at the xMII. The value PMAC indicates that the TS\_TX.indication primitive is associated with a packet that has that an SMD-S value has been detected at the xMII (see Table 99–1). The MM parameter is not provided when MAC Merge sublayer is not instantiated.~~

When MM = PMAC, the TS\_TX.indication is not generated for continuation fragments.

The PCS dynamic path data delay (PDPDD) is an optional parameter that supports high accuracy dynamic transmit path data delay calculations. It provides a value ranging from –32 768 to +32 767 indicating the number of bit times (see 1.4.215) of dynamic transmit path data delay the DDMP experiences in the PCS within the PHY. A positive value represents an addition to the mean of the maximum and minimum PCS transmit path data delay values given by the PCS transmit path data delay registers (see 45.2.3.68). A negative value represents a reduction from the mean of the maximum and minimum PCS transmit path data delay values given by the PCS transmit path data delay registers. The PDPDD value is conveyed from the PHY to the gRS by the optional TX\_NUM\_BIT\_CHANGE<15:0> signals. See 90.5.3.

NOTE—When using the half-duplex mode of operation, multiple TS\_TX indications may be produced for a single MA\_DATA.request as a result of collisions on the media.

### 90.4.3.1.2 Condition for generation

*Change 90.4.3.1.2 and 90.4.3.1.3 as follows:*

This primitive is generated by the gRS sublayer in response to detection of a valid ~~SFD~~ DDMP in the data stream transmitted across the xMII transmit signals. Specific conditions for generation of this primitive are described in 90.5.1.

### 90.4.3.1.3 Effect of receipt

The receipt of this primitive by the TimeSync Client is ~~undefined~~ beyond the scope of this standard.

### 90.4.3.2 TS\_RX.indication primitive

*Change 90.4.3.2 as follows:*

This primitive defines the transfer of the indication of an event between the gRS and the TimeSync Client. When asserted, the TimeSync Client is notified that a valid DDMP was detected by the gRS TS\_DDMP\_Detect\_RX function (see 90.5.2) in the xMII receive signals.

#### 90.4.3.2.1 Semantics

*Change 90.4.3.2.1 (as amended by IEEE Std 802.3de-2022) as follows:*

The semantics of the primitive are as follows:

TS\_RX.indication(~~SFD~~, ~~DDMP~~, MM, PDPDD)

~~The SFD parameter can take only one possible value, DETECTED. When asserted (SFD = DETECTED), the TimeSync Client is notified that a valid SFD was detected by the gRS sublayer TS\_SFD\_Detect\_RX function (see 90.5.2) in the xMII receive signals.~~

The DDMP parameter can take one of two possible values, SFD and FIRST\_SYMBOL. The value SFD indicates that the TS\_RX.indication primitive was generated as the result of the beginning of an SFD being transferred across the receive path of the xMII. The value FIRST\_SYMBOL indicates that the TS\_RX.indication primitive was generated as the result of the beginning of the first symbol after an SFD being transferred across the receive path of the xMII.

The information given by the DDMP parameter may be used by a TimeSync Client to get bit time resolution for the timestamp capture for the DDMP in the xMII word.

The use of the beginning of the SFD, or the beginning of the first symbol after the SFD, as the DDMP requires consistent configuration of both the gRS and the PHY (see 45.2.3.69a and 45.2.5.31) for correct operation.

The MM parameter is mandatory when the MAC Merge sublayer (see [Clause 99](#)) is instantiated. The MM parameter, when present, can take one of two possible values, ~~i.e.,~~ PMAC or EMAC. The value EMAC indicates that the TS\_RX.indication primitive is associated with a packet that has an SMD-E (SFD) value has been detected at the xMH. The value PMAC indicates that the TS\_RX.indication primitive is associated with a packet that has that an SMD-S value has been detected at the xMH (see [Table 99–1](#)). ~~The MM parameter is not provided when MAC Merge sublayer is not instantiated.~~

When MM = PMAC, the TS\_RX.indication is not generated for continuation fragments.

The PCS dynamic path data delay (PDPDD) is an optional parameter that supports high accuracy dynamic receive path data delay calculations. It provides a value ranging from –32 768 to +32 767 indicating the number of bit times (see [1.4.2.15](#)) of dynamic receive path data delay the DDMP experiences in the PCS within the PHY. A positive value represents an addition to the mean of the maximum and minimum PCS receive path data delay values given by the PCS receive path data delay registers (see 45.2.3.69). A negative value represents a reduction from the mean of the maximum and minimum PCS receive path data delay values given by the PCS receive path data delay registers. The PDPDD value is conveyed from the PHY to the gRS by the optional RX\_NUM\_BIT\_CHANGE<15:0> signals. See 90.5.4.

NOTE—When using the half-duplex mode of operation, multiple TS\_RX indications may be produced for a single packet as a result of collisions on the media.

#### 90.4.3.2.2 Condition for generation

*Change 90.4.3.2.2 and 90.4.3.2.3 as follows:*

This primitive is generated by the gRS sublayer in response to detection of a valid ~~SFD-DDMP~~ in the data stream received across the xMII receive signals. Specific conditions for generation of this primitive are described in 90.5.2.

#### 90.4.3.2.3 Effect of receipt

The receipt of this primitive by the TimeSync Client is ~~undefined~~ beyond the scope of this standard.

### 90.5 generic Reconciliation Sublayer (gRS)

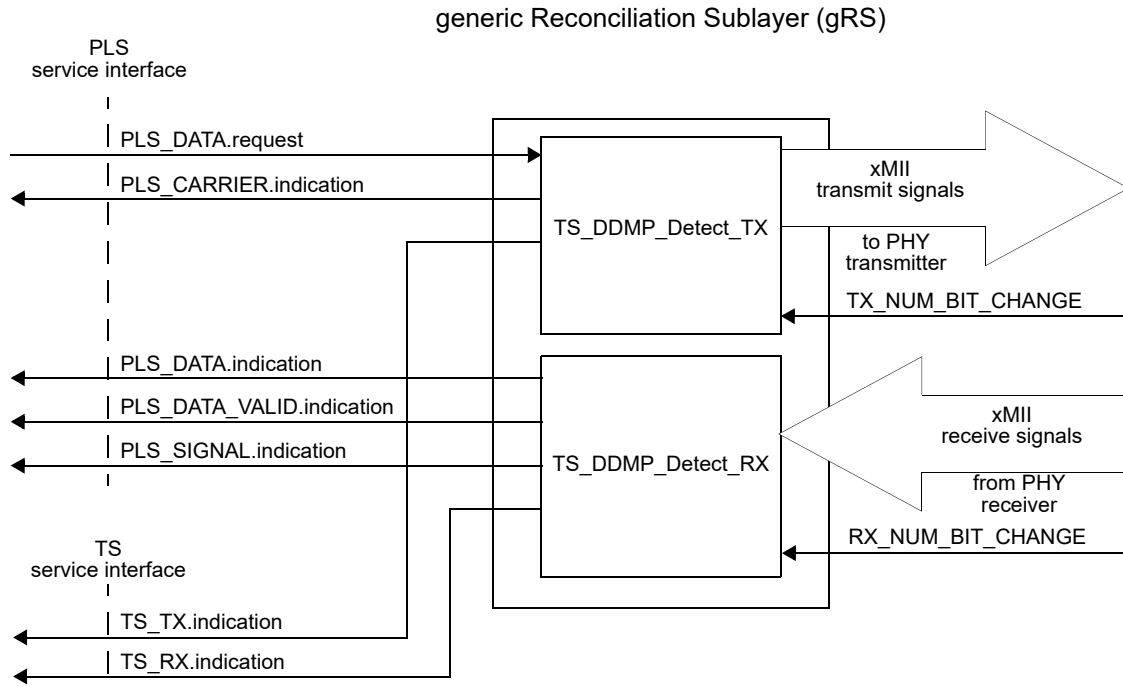
*Change the text of 90.5 as follows:*

Within the scope of this clause, the term generic Reconciliation Sublayer (gRS) is used to denote any IEEE 802.3 Reconciliation Sublayer (RS) used to interface a MAC with any PHY supporting the TimeSync capability through the xMII. The behavior of the gRS depends on whether a MAC Merge sublayer (see [Clause 99](#)) is instantiated above the gRS.

For the purpose of the TimeSync capabilities, two new functions are defined in this subclause, namely ~~TS\_SFD\_Detect\_TX~~ TS\_DDMP\_Detect\_TX (see 90.5.1) and ~~TS\_SFD\_Detect\_RX~~ TS\_DDMP\_Detect\_RX (see 90.5.2), which are responsible for generation of the TS\_TX.indication and TS\_RX.indication service primitives, as defined in 90.4. Figure 90–2 presents the ~~TS\_SFD\_Detect\_TX~~ TS\_DDMP\_Detect\_TX and ~~TS\_SFD\_Detect\_RX~~ TS\_DDMP\_Detect\_RX functions and their location within the gRS sublayer.

In addition, an optional bundle of 16 logical transmit signals (TX\_NUM\_BIT\_CHANGE<15:0>) and 16 logical receive signals (RX\_NUM\_BIT\_CHANGE<15:0>) may be output from the PHY to the gRS. These signals provide the dynamic path data delay information that is used to generate the PDPDD parameter passed to the TimeSync Client in the TS\_TX.indicate and TS\_RX.indicate primitives, respectively, to enable the calculation of higher accuracy path data delay values.

Replace Figure 90–2 with the following figure:



**Figure 90–2—TS\_DDMP\_Detect\_TX and TS\_DDMP\_Detect\_RX functions within the generic Reconciliation Sublayer (gRS)**

Change the title and text of 90.5.1 as follows:

#### 90.5.1 ~~TS\_SFD\_Detect\_TX~~ TS\_DDMP\_Detect\_TX function

The ~~TS\_SFD\_Detect\_TX~~ TS\_DDMP\_Detect\_TX function observes the xMII transmit signals.

When the MAC Merge sublayer is not instantiated, the ~~TS\_SFD\_Detect\_TX~~ TS\_DDMP\_Detect\_TX function detects the occurrence of the ~~DDMP~~ Start Frame Delimiter (SFD, see 3.1.1 and 3.2.2) in compliance with the specifications of the given type of the instantiated xMII. The service primitive across the TSSI, i.e., TS\_TX.indication primitive shall be generated only when the ~~SFD~~ DDMP is detected on the transmit signals of the xMII (~~SFD-DETECTED~~).

When the MAC Merge sublayer is instantiated and the beginning of the SFD is selected as the DDMP, the ~~TS\_SFD\_Detect\_TX~~ TS\_DDMP\_Detect\_TX function detects the occurrence of the beginning of SMD-E or SMD-S. The TS\_TX.indication primitive shall be generated only when the beginning of the SMD-E for an express packet or the beginning of the SMD-S for a preemptable packet (see 99.3.3) is detected on the transmit signals of the xMII. The value of the MM parameter of the TS\_TX.indication primitive shall be set to EMAC if an SMD-E is detected and to PMAC if an SMD-S is detected. ~~Start mPacket Delimiter for an express packet or preemptable packet start (SMD-E or SMD-S, see 99.3.3) in compliance with the specifications of the given type of the instantiated xMII. The service primitive across the TSSI, i.e., TS\_TX.indication, shall be generated only when the SMD-E or SMD-S is detected on the transmit signals of the xMII (SFD-DETECTED). The value of MM shall indicate whether an SMD-E (MM=EMAC) or an SMD-S (MM=PMAC) was detected.~~

When the MAC Merge sublayer is instantiated and the beginning of the first symbol after the SFD is selected as the DDMP, the TS\_DDMP\_Detect\_TX function detects the occurrence of the beginning of the

first symbol after SMD-E or SMD-S. The TS\_TX.indication primitive shall be generated only when the beginning of the first symbol after the SMD-E for an express packet or the beginning of the first symbol after the SMD-S for a preemptable packet (see 99.3.3) is detected on the transmit signals of the xMII. The value of the MM parameter of the TS\_TX.indication primitive shall be set to EMAC if an SMD-E preceded the DDMP and to PMAC if an SMD-S preceded the DDMP.

*Change the title and text of 90.5.2 as follows:*

### **90.5.2 TS\_SFD\_Detect\_RX-TS\_DDMP\_Detect\_RX function**

The TS\_SFD\_Detect\_RX-TS\_DDMP\_Detect\_RX function observes the xMII receive signals.

When the MAC Merge sublayer is not instantiated, the TS\_SFD\_Detect\_RX-TS\_DDMP\_Detect\_RX function detects the occurrence of the DDMP—Start Frame Delimiter (SFD, see 3.1.1 and 3.2.2) in compliance with the specifications of the given type of the instantiated xMII. The service primitive across the TSSI, i.e., TS\_RX.indication; primitive shall be generated only when the SFD-DDMP is detected on the receive signals of the xMII (SFD-DETECTED).

When the MAC Merge sublayer is instantiated and the beginning of the SFD is selected as the DDMP, the TS\_SFD\_Detect\_RX-TS\_DDMP\_Detect\_RX function detects the occurrence of the beginning of SMD-E or SMD-S. The TS\_RX.indication primitive shall be generated only when the beginning of the SMD-E for an express packet or the beginning of the SMD-S for a preemptable packet is detected on the receive signals of the xMII. The value of the MM parameter of the TS\_RX.indication primitive shall be set to EMAC if an SMD-E is detected and to PMAC if an SMD-S is detected. ~~Start mPacket Delimiter for an express packet or preemptable packet start (SMD-E or SMD-S, see 99.3.3) in compliance with the specifications of the given type of the instantiated xMII. The service primitive across the TSSI, i.e., TS\_RX.indication, shall be generated only when the SMD-E or SMD-S is detected on the transmit signals of the xMII (SFD-DETECTED). The value of MM shall indicate whether an SMD-E (MM-EMAC) or an SMD-S (MM-PMAC) was detected.~~

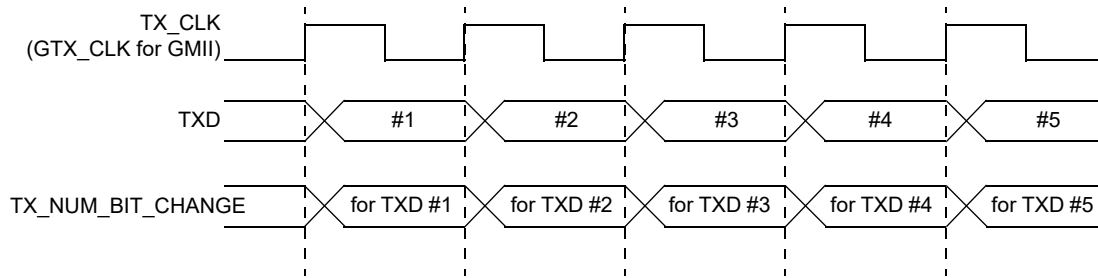
When the MAC Merge sublayer is instantiated and the beginning of the first symbol after the SFD is selected as the DDMP, the TS\_DDMP\_Detect\_RX function detects the occurrence of the beginning of the first symbol after SMD-E or SMD-S. The TS\_RX.indication primitive shall be generated only when the beginning of the first symbol after the SMD-E for an express packet or the beginning of the first symbol after the SMD-S for a preemptable packet (see 99.3.3) is detected on the receive signals of the xMII. The value of the MM parameter of the TS\_RX.indication primitive shall be set to EMAC if an SMD-E preceded the DDMP and to PMAC if an SMD-S preceded the DDMP.

*Insert new subclauses 90.5.3 and 90.5.4 after 90.5.2 as follows:*

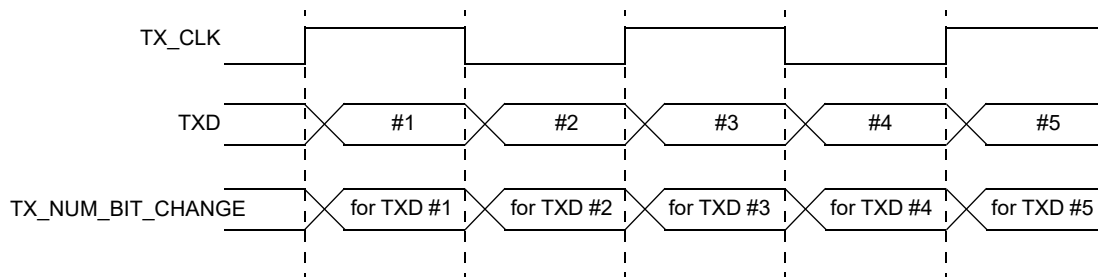
### **90.5.3 Dynamic transmit path data delay**

The optional dynamic transmit path data delay information supports the calculation of higher accuracy dynamic transmit path data delay values by the TimeSync Client. It is provided as a bundle of 16 logical signals (TX\_NUM\_BIT\_CHANGE<15:0>). These logical signals are intended for use with an intra-chip interface. A physical instantiation of these logical signals is not defined. The dynamic transmit path data delay information is sourced by the PHY for each active TX\_CLK (GTX\_CLK for GMII) edge upon which transmit data is transferred from the gRS to the PHY and provides a value ranging from -32 768 to +32 767 in two's complement format. The value reports the number of bit times of dynamic transmit path data delay that are experienced by the data transferred from the gRS to the PHY, relative to the mean PHY transmit path data delay (see 45.2.3.68), due to actions such as alignment marker insertion, codeword marker insertion, and/or idle rate adaptation insertion/removal. The relationship between TX\_CLK, TXD, and TX\_NUM\_BIT\_CHANGE is illustrated in Figure 90–2a for an xMII that is active only on the rising

TX\_CLK edge. The relationship between TX\_CLK, TXD, and TX\_NUM\_BIT\_CHANGE is illustrated in Figure 90–2b for an xMII that is active on both rising and falling TX\_CLK edges.



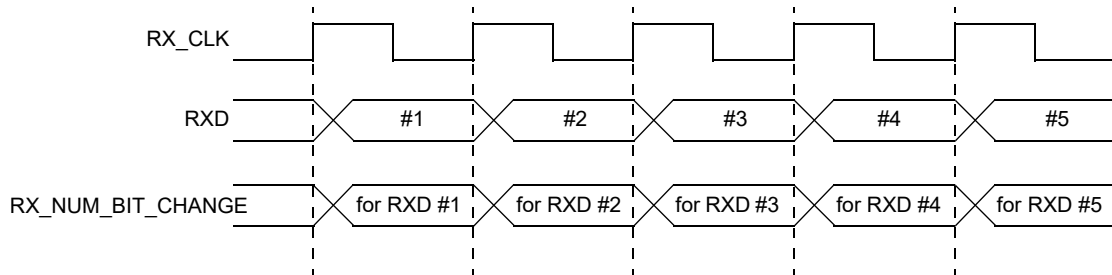
**Figure 90–2a—Relationship between TXD and TX\_NUM\_BIT\_CHANGE for active rising TX\_CLK edge**



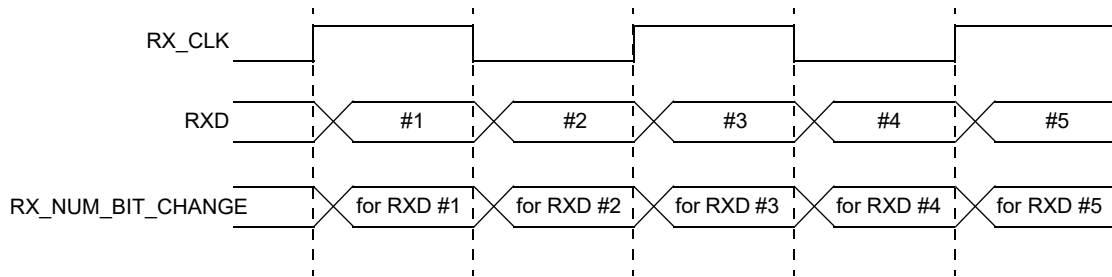
**Figure 90–2b—Relationship between TXD and TX\_NUM\_BIT\_CHANGE for active rising and falling TX\_CLK edges**

#### 90.5.4 Dynamic receive path data delay

The optional dynamic receive path data delay information supports the calculation of higher accuracy dynamic receive path data delay values by the TimeSync Client. It is provided as a bundle of 16 logical signals (RX\_NUM\_BIT\_CHANGE<15:0>). These logical signals are intended for use with an intra-chip interface. A physical instantiation of these logical signals is not defined. The dynamic receive path data delay information is sourced by the PHY for each RX\_CLK (GRX\_CLK for GMII) edge upon which receive data is transferred from the PHY to the gRS and provides a value ranging from –32 768 to +32 767 in two’s complement format. The value reports the number of bit times of dynamic receive path data delay that are experienced by the data transferred from the PHY to the gRS, relative to the mean PHY receive path data delay (see 45.2.3.69), due to actions such as alignment marker removal, codeword marker removal, and/or idle rate adaptation insertion/removal. The relationship between RX\_CLK, RXD, and RX\_NUM\_BIT\_CHANGE is illustrated in Figure 90–2c for an xMII that is active only on the rising RX\_CLK edge. The relationship between the RX\_CLK, RXD, and RX\_NUM\_BIT\_CHANGE is illustrated in Figure 90–2d for an xMII that is active on both rising and falling RX\_CLK edges.



**Figure 90-2c—Relationship between RXD and RX\_NUM\_BIT\_CHANGE for active rising RX\_CLK edge**



**Figure 90-2d—Relationship between RXD and RX\_NUM\_BIT\_CHANGE for active rising and falling RX\_CLK edges**

## 90.6 Overview of management features

*Change 90.6 as follows:*

Clause 30 describes the management functions for any of the IEEE Std 802.3 compliant PHYs. Objects defined in Clause 30 for the support of TimeSync capability are summarized as follows:

- oTimeSync managed object class, as defined in 30.13.1
- ~~aTimeSyncCapabilityNsTX, and aTimeSyncCapabilityNsRX, aTimeSyncCapabilitySubNsTX, aTimeSyncCapabilitySubNsRX, aTimeSyncCapabilityDdmp, aTimeSyncCapabilityMultiplePcsLane, and aTimeSyncCapabilityDynamicPathDataDelay managed objects, as defined in 30.13.1.1, 30.13.1.2, 30.13.1.7, 30.13.1.8, 30.13.1.13, 30.13.1.15, and 30.13.1.16, respectively, reflecting the status of a series of MDIO capability registers (1.1800, 2.1800, 3.1800, 4.1800, 5.1800, and 6.1800), as defined in 30.13.1.1 and 30.13.1.2, respectively~~
- aTimeSyncDelayNsTXmax and aTimeSyncDelayNsTXmin managed objects, representing the aggregate value of the series of transmit path data delay registers present in the instantiated MDIO registers (for each MMD, in case of multiple instances) 1.1801 through 1.1804, 2.1801 through 2.1804, 3.1801 through 3.1804, 4.1801 through 4.1804, 5.1801 through 5.1804, and 6.1801 through 6.1804, as defined in 30.13.1.3 and 30.13.1.4, ~~respectively~~
- aTimeSyncDelayNsRXmax and aTimeSyncDelayNsRXmin managed objects, representing the aggregate value of the series of receive path data delay registers present in the instantiated MDIO registers (for each MMD, in case of multiple instances) 1.1805 through 1.1808, 2.1805 through 2.1808, 3.1805 through 3.1808, 4.1805 through 4.1808, 5.1805 through 5.1808, and 6.1805 through 6.1808, as defined in 30.13.1.5 and 30.13.1.6, ~~respectively~~

- aTimeSyncDelaySubNsTXmax and aTimeSyncDelaySubNsTXmin managed objects, representing the aggregate value of the series of transmit path data delay registers present in the instantiated MDIO registers (for each MMD, in case of multiple instances) 1.1809, 1.1810, 2.1809, 2.1810, 3.1809, 3.1810, 4.1809, 4.1810, 5.1809, 5.1810, 6.1809, and 6.1810, as defined in 30.13.1.9 and 30.13.1.10.
- aTimeSyncDelaySubNsRXmax and aTimeSyncDelaySubNsRXmin managed objects, representing the aggregate value of the series of receive path data delay registers present in the instantiated MDIO registers (for each MMD, in case of multiple instances) 1.1811, 1.1812, 2.1811, 2.1812, 3.1811, 3.1812, 4.1811, 4.1812, 5.1811, 5.1812, 6.1811, and 6.1812, as defined in 30.13.1.11 and 30.13.1.12.
- aTimeSyncDdmpSelection managed object, used to select or get the DDMP configuration state, as either the beginning of the SFD or the beginning of the first symbol after the SFD, via the instantiated MDIO registers 3.1813.13 and 5.1813.13, as defined in 30.13.1.14.

The Management Data Input/Output (MDIO) capability described in Clause 45 defines several variables that provide TimeSync status information for the PMD, as shown in Table 90–1:

TimeSync status, capability, and configuration information for the PHY, as shown in Table 90–1, may be accessible through the management interface defined in Clause 45, or equivalent.

**Table 90–1—Summary of TimeSync features in Clause 45**

| Register                                    | Name                                       | Reference         |
|---------------------------------------------|--------------------------------------------|-------------------|
| 1.1800                                      | TimeSync PMA/PMD capability register       | 45.2.1.175        |
| 1.1801 through 1.1804,<br>1.1809 and 1.1810 | TimeSync PMA/PMD transmit path data delay  | 45.2.1.176        |
| 1.1805 through 1.1808,<br>1.1811 and 1.1812 | TimeSync PMA/PMD receive path data delay   | 45.2.1.177        |
| 2.1800                                      | TimeSync WIS capability register           | 45.2.2.20         |
| 2.1801 through 2.1804,<br>2.1809 and 2.1810 | TimeSync WIS transmit path data delay      | 45.2.2.21         |
| 2.1805 through 2.1808,<br>2.1811 and 2.1812 | TimeSync WIS receive path data delay       | 45.2.2.22         |
| 3.1800                                      | TimeSync PCS capability register           | 45.2.3.67         |
| 3.1801 through 3.1804,<br>3.1809 and 3.1810 | TimeSync PCS transmit path data delay      | 45.2.3.68         |
| 3.1805 through 3.1808,<br>3.1811 and 3.1812 | TimeSync PCS receive path data delay       | 45.2.3.69         |
| 3.1813                                      | <u>TimeSync PCS configuration register</u> | <u>45.2.3.69a</u> |
| 4.1800                                      | TimeSync PHY XS capability register        | 45.2.4.28         |
| 4.1801 through 4.1804,<br>4.1809 and 4.1810 | TimeSync PHY XS transmit path data delay   | 45.2.4.29         |
| 4.1805 through 4.1808,<br>4.1811 and 4.1812 | TimeSync PHY XS receive path data delay    | 45.2.4.30         |
| 5.1800                                      | TimeSync DTE XS capability register        | 45.2.5.28         |

**Table 90–1—Summary of TimeSync features in Clause 45 (*continued*)**

| Register                                    | Name                                          | Reference        |
|---------------------------------------------|-----------------------------------------------|------------------|
| 5.1801 through 5.1804_<br>5.1809 and 5.1810 | TimeSync DTE XS transmit path data delay      | 45.2.5.29        |
| 5.1805 through 5.1808_<br>5.1811 and 5.1812 | TimeSync DTE XS receive path data delay       | 45.2.5.30        |
| 5.1813                                      | <u>TimeSync DTE XS configuration register</u> | <u>45.2.5.31</u> |
| 6.1800                                      | TimeSync TC capability register               | 45.2.6.14        |
| 6.1801 through 6.1804_<br>6.1809 and 6.1810 | TimeSync TC transmit path data delay          | 45.2.6.15        |
| 6.1805 through 6.1808_<br>6.1811 and 6.1812 | TimeSync TC receive path data delay           | 45.2.6.16        |

*Change the title and first paragraph of 90.7 as follows:*

## **90.7 Path dData delay measurement**

~~The TimeSync capability requires measurement of data delay in the transmit and receive paths, as shown in Figure 90–3. The transmit path data delay is measured from the beginning of the SFD at the xMII input to the beginning of the SFD at the MDI output. The receive path data delay is measured from the beginning of the SFD at the MDI input to the beginning of the SFD at the xMII output.~~

The TimeSync Client may use the egress and ingress times captured at the xMII and the transmit and receive path data delay measurements, as shown in Figure 90–3, to estimate the corresponding egress and ingress times at the MDI. The path data delay measurements are based on the selected DDMP. The DDMP shall be either the beginning of the SFD or the beginning of the first symbol after the SFD (see 45.2.3.69a and 45.2.5.31). The choice of the DDMP is implementation-dependent but requires consistent configuration of both the gRS and the PHY for correct operation.

Replace Figure 90–3 with the following figure:

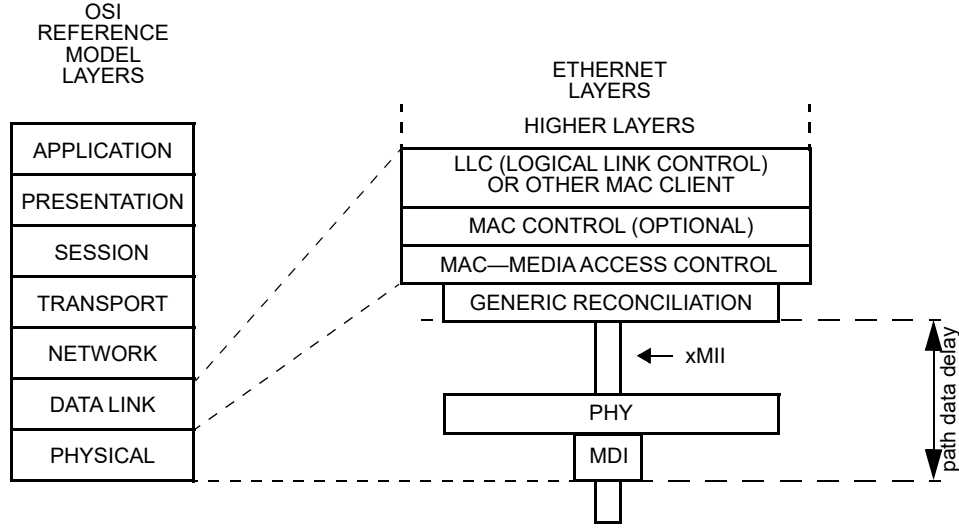


Figure 90–3—Path data delay measurement

Change the second to fourth paragraph of 90.7 (including three notes) as follows:

For a PHY that includes an FEC function, the transmit and receive path data delays may show significant variation depending upon the position of the SFD within the FEC block. However, since the variation due to this effect in the transmit path is expected to be compensated by the inverse variation in the receive path, it is recommended that the transmit and receive path data delays be reported as if the SFD is at the start of the FEC block.

The receiver of a multi-lane PHY is expected to include a buffer to compensate for skew between the lanes. This buffer selectively delays each lane such that the lanes are aligned at the buffer output. The earliest arriving lane experiences the most delay through the buffer and the latest arriving lane experiences the least delay through the buffer. The receive path data delay for a multi-lane PHY is reported as if the beginning of the SFD arrived at the MDI input on the lane with the smallest buffer delay.

NOTE 1—The measured data delay in either the transmit or receive path should not include delay resulting from any length of the medium, regardless of the type of the medium used by the given PHY. It is observed that even if any medium-imposed delay is included in the measured data delay, it does not affect the operation of the TimeSync functions and only shifts the timing reference point farther from the MDI.

The obtained data delay measurement shall be reported in the form of a quartet of values; the maximum transmit data delay, the minimum transmit data delay, the maximum receive data delay, and the minimum receive data delay, as defined for the oTimeSync managed object class (30.13.1).

NOTE 2—The data delay values represent only the data delay in the PHY sublayers. The TimeSync Client may need to adjust for delays within the GRS. For example, the TimeSync Client may need to subtract the value of the delay associated with the TS\_SFD\_Detect\_TX function from the sum of the minimum transmit data delay values reported by individual MMD(s). Likewise, the TimeSync Client may need to add the value of the delay associated with the TS\_SFD\_Detect\_RX function to the sum of the maximum receive data delay values reported by individual MMD(s).

NOTE 3—For PHYs that are specified with an FEC sublayer separate from the PCS, the data delay for the FEC sublayer should be included in either the PCS delay registers or the PMA/PMD delay registers of the MMD in which the FEC sublayer is implemented, but not in both.

NOTE 1—It is recommended that the beginning of the first symbol after the SFD is used as the DDMP since this matches the measurement points specified in IEEE Std 802.1AS [B39] and IEEE Std 1588 [B41]. The use of the

beginning of the SFD as the DDMP can impact the accuracy that can be achieved by a time synchronization protocol using this TSSI. Further information can be found in Annex 90A.

The transmit path data delay is defined as the time it takes for a packet's DDMP to pass from the xMII input to the MDI output. The receive path data delay is defined as the time it takes for a packet's DDMP to pass from the MDI input to the xMII output.

To obtain the egress or ingress time of the DDMP of a packet at the MDI, the TimeSync Client may adjust the egress or ingress time that it captured for the DDMP of the packet at the xMII by the path data delay, if available (see Figure 90–3) and the dynamic path data delay, if supplied (see 90.4.2). The accuracy of the calculated egress or ingress time at the MDI is therefore dependent on the accuracy of the transmit and receive path data delay and dynamic path data delay values.

NOTE 2—The measured path data delay in either the transmit or receive path does not include delay resulting from any length of the medium, regardless of the type of the medium used by the given PHY. It is observed that even if any medium-imposed delay is included in the measured path data delay, it does not affect the operation of the TimeSync functions and only shifts the timestamp generation reference plane for one end of the path data delay measurement farther from the MDI.

To support the calculation of the path data delay, each MMD within a PHY may provide up to two quartets of values, one quartet provides the integer nanoseconds portion, the other quartet provides the fractional nanoseconds portion. Each quartet consists of two pairs of values, the maximum transmit path data delay and the minimum transmit path data delay, and the maximum receive path data delay and the minimum receive path data delay, as defined for the oTimeSync managed object class (30.13.1). The provision by an MMD of either pair of values for either portion is independently optional. A single quartet of values for the PHY path data delay may be obtained by summing together the values, if available, of each corresponding member of both quartets for each MMD. The uncertainty of the transmit and receive path data delays of the corresponding sublayer may be determined from this quartet of values. The path data delay error of the sublayer can be minimized by using the mean of its maximum and minimum path data delay values as its path data delay value.

NOTE 3—The path data delay values represent only the path data delay in the PHY sublayers. The TimeSync Client might also compensate for implementation-specific delays that are not accounted for by the path data delay values reported by the PHY sublayer managed objects from 30.13.1 and by the PDPDD parameters from 90.4.3. The handling of these implementation-specific delays is beyond the scope of this standard.

*Insert new subclauses 90.7.1, 90.7.2, and 90.7.3 at the end of 90.7 as follows:*

#### **90.7.1 FEC and PCS lane distribution functions**

For a PHY that includes an FEC and/or a PCS lane distribution function, the transmit and receive path data delays can show significant variation depending on how the packet's DDMP aligns to an FEC codeword and/or to a PCS lane distribution sequence. However, since the variation due to this effect in the transmit path is expected to be compensated by the inverse variation in the receive path, it is recommended that the transmit and receive path data delays be reported as if the DDMP is at the start of the FEC codeword and/or at the start of the PCS lane distribution sequence (when the multilane ability (3.1800.11) bit is read as one, see 45.2.3.67.2). For example, for many PHYs with both FEC and PCS lane distribution, the start of an FEC codeword necessarily coincides with the start of a PCS lane distribution sequence periodically during their respective processions. This results in the maximum FEC and the maximum PCS lane distribution delays being allocated to the PHY transmitter and the minimum FEC and the minimum PCS lane distribution delays being allocated to the PHY receiver. See 90A.7 for more information on the cascading delays of multiple functions.

NOTE—For PHYs that are specified with an FEC sublayer separate from the PCS, the path data delay for the FEC sublayer may be included in either the PCS delay registers or the PMA/PMD delay registers of the MMD in which the FEC sublayer is implemented, but not in both.

### 90.7.2 Alignment marker, codeword marker, and idle insertion/removal functions

For a PHY that inserts alignment markers or codeword markers and/or performs rate adaptation (e.g., adds/removes idles), the transmit path data delay measurement starting event (when the DDMP passes the xMII input) should be adjusted to account for the alignment marker or codeword marker insertion and any idle insertion/removal that occurs in the PHY (between the xMII input and the MDI output). Based on this adjustment, the result is a transmit path data delay measurement that appears as if the alignment marker or codeword marker insertion and any idle insertion/removal had been performed before the transmit xMII. The PHY provides its transmit path data delay variance information to the TimeSync Client via the PDPDD parameter in the TS\_TX.indication primitive.

For a PHY that removes alignment markers or codeword markers and/or performs rate adaptation (e.g., adds/removes idles), the receive path data delay measurement ending event (when the DDMP passes the xMII output) should be adjusted to account for alignment marker or codeword marker removal and any idle insertion/removal that occurs in the PHY (between the MDI input and xMII output). Based on this adjustment, the result is a receive path data delay measurement that appears as if the alignment marker or codeword marker removal and any idle insertion/removal had been performed after the receive xMII. The PHY provides its receive path data delay variance information to the TimeSync Client via the PDPDD parameter in the TS\_RX.indication primitive.

The dynamic delay variance of alignment marker, codeword marker, or idle insertion or removal is not to be included in the TimeSync PCS transmit path data delay or the TimeSync PCS receive path data delay registers.

NOTE—When TX\_NUM\_BIT\_CHANGE and RX\_NUM\_BIT\_CHANGE are not available (e.g., over physical interfaces such as instantiated xMII or AUI), it is recommended to avoid insertion and removal of idles, alignment markers, and codeword markers in the sublayers below the xMII/AUI, when possible, to reduce timestamping accuracy impairments (see Annex 90A).

### 90.7.3 Lane skew

Lane skew is possible on a transmitter with multiple PCS and PMA/PMD lanes when these lanes have different static latencies such that their alignment markers appear staggered as they depart the device at the MDI output. Since transmit skew in series with medium skew is not strictly additive, transmit skew can contribute to time synchronization error by obscuring the actual latency of the medium. Transmit skew is expected to be minimized, ideally to zero, representing an ideal case for the accuracy of a TimeSync Client. If the transmit skew is not zero, then it is recommended that the transmit path data delay for a PHY with multiple lanes be reported as if the DDMP departed the MDI output on an imaginary lane whose departure time is the midpoint between those of the first-departing and last-departing lanes. This has the effect of centering any timestamp error in the middle of the skew window.

The receiver of a PHY with multiple lanes is expected to include a buffer to compensate for skew between the lanes. This buffer selectively delays each lane such that the lanes are aligned at the buffer output. The earliest arriving lane experiences the most delay through the buffer and the latest arriving lane experiences the least delay through the buffer. The receive path data delay for a PHY with multiple lanes is reported as if the DDMP arrived at the MDI input on the lane with the smallest buffer delay.

NOTE—For specific cases when transmit skew is strictly additive to any medium skew, it is appropriate to report the transmit path data delay as if the DDMP departed the MDI output on the last-departing lane.

## 90.8 Protocol implementation conformance statement (PICS) proforma for Clause 90, Ethernet support for time synchronization protocols<sup>7</sup>

### 90.8.3 TSSI indication

Change the table in 90.8.3 as follows:

| Item               | Feature                                                                                                                                                    | Subclause         | Value/Comment                                                                                                                                                                                                                                                                                                                                     | Status          | Support                                  |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------------------------------------|
| *MM                | MAC Merge                                                                                                                                                  | 90.5              | A MAC Merge sublayer is instantiated above gRS.                                                                                                                                                                                                                                                                                                   | O               | Yes [ ]<br>No [ ]                        |
| TS_TX              | TS_TX.indication generation                                                                                                                                | 90.5.1            | TS_TX.indication is generated only when the <del>SFD-DDMP</del> is detected on the transmit signals of the xMII.                                                                                                                                                                                                                                  | <del>MM:M</del> | Yes [ ]<br><del>N/A [ ]</del>            |
| <del>TS_RX</del>   | <del>TS_RX.indication generation</del>                                                                                                                     | <del>90.5.2</del> | <del>TS_RX.indication is generated only when the SFD is detected on the receive signals of the xMII.</del>                                                                                                                                                                                                                                        | <del>MM:M</del> | <del>Yes [ ]</del><br><del>N/A [ ]</del> |
| TS_T2<br>TS_TX_MM1 | <del>TS_TX.indication generation with MAC Merge</del><br>TS_TX.indication generation. MAC Merge sublayer, the beginning of the SFD is selected as the DDMP | 90.5.1            | <del>TS_TX.indication is generated only when SMD-E or SMD-S is detected on the transmit signals of the xMII.</del><br>The TS_TX.indication primitive is generated only when the beginning of the SMD-E for an express packet or the beginning of the SMD-S for a preemptable packet (see 99.3.3) is detected on the transmit signals of the xMII. | MM:M            | Yes [ ]<br>N/A [ ]                       |
| TS_TX_MM2          | TS_TX.indication generation. MAC Merge sublayer, the beginning of first symbol after the SFD is selected as the DDMP                                       | 90.5.1            | The TS_TX.indication primitive is generated only when the beginning of the first symbol after the SMD-E for an express packet or the beginning of the first symbol after the SMD-S for a preemptable packet (see 99.3.3) is detected on the transmit signals of the xMII.                                                                         | MM:M            | Yes [ ]<br>N/A [ ]                       |
| TS_T3<br>TS_TX_MM3 | <del>MM parameter for TS_TX</del><br>The value of MM parameter of the TS_TX.indication primitive, the beginning of the SFD is selected as the DDMP         | 90.5.1            | <del>Value of MM parameter indicates whether an SMD-E or SMD-S is detected.</del><br>When the MAC Merge sublayer is instantiated and the beginning of the SFD is selected as the DDMP, the value of the MM parameter of the TS_TX.indication primitive is set to EMAC if an SMD-E is detected and to PMAC if an SMD-S is detected.                | MM:M            | Yes [ ]<br>N/A [ ]                       |

<sup>7</sup>Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

| Item                             | Feature                                                                                                                                      | Subclause     | Value/Comment                                                                                                                                                                                                                                                                                                               | Status      | Support                          |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------------------|
| <u>TS_TX_MM4</u>                 | <u>The value of MM parameter of the TS_TX.indication primitive, the beginning of the symbol after the SFD is selected as the DDMP</u>        | <u>90.5.1</u> | <u>When the MAC Merge sublayer is instantiated and the beginning of the symbol after SFD is selected as the DDMP, the value of the MM parameter of the TS_TX.indication primitive is set to EMAC if an SMD-E preceded the DDMP and to PMAC if an SMD-S preceded the DDMP.</u>                                               | <u>MM:M</u> | <u>Yes [ ]</u><br><u>N/A [ ]</u> |
| <u>TS_RX</u>                     | <u>TS_RX.indication generation</u>                                                                                                           | <u>90.5.2</u> | <u>TS_RX.indication is generated only when the DDMP is detected on the receive signals of the xMII.</u>                                                                                                                                                                                                                     | <u>M</u>    | <u>Yes [ ]</u>                   |
| <u>TS_R2</u><br><u>TS_RX_MM1</u> | <u>TS_RX.indication generation with MAC Merge sublayer, the beginning of the SFD is selected as the DDMP</u>                                 | <u>90.5.2</u> | <u>TS_RX.indication is generated only when SMD-E or SMD-S is detected on the receive signals of the xMII. The TS_RX.indication primitive is generated only when the beginning of the SMD-E for an express packet or the beginning of the SMD-S for a preemptable packet is detected on the receive signals of the xMII.</u> | <u>MM:M</u> | <u>Yes [ ]</u><br><u>N/A [ ]</u> |
| <u>TS_RX_MM2</u>                 | <u>TS_RX.indication generation, MAC Merge sublayer, the beginning of first symbol after the SFD is selected as the DDMP</u>                  | <u>90.5.2</u> | <u>The TS_RX.indication primitive is generated only when the beginning of the first symbol after the SMD-E for an express packet or the beginning of the first symbol after the SMD-S for a preemptable packet (see 99.3.3) is detected on the receive signals of the xMII.</u>                                             | <u>MM:M</u> | <u>Yes [ ]</u><br><u>N/A [ ]</u> |
| <u>TS_R3</u><br><u>TS_RX_MM3</u> | <u>MM parameter for TS_RX. The value of MM parameter of the TS_RX.indication primitive, the beginning of the SFD is selected as the DDMP</u> | <u>90.5.2</u> | <u>Value of MM parameter indicates whether an SMD-E or SMD-S is detected. When the MAC Merge sublayer is instantiated and the beginning of the SFD is selected as the DDMP, the value of the MM parameter of the TS_RX.indication primitive is set to EMAC if an SMD-E is detected and to PMAC if an SMD-S is detected.</u> | <u>MM:M</u> | <u>Yes [ ]</u><br><u>N/A [ ]</u> |
| <u>TS_RX_MM4</u>                 | <u>The value of MM parameter of the TS_RX.indication primitive, the beginning of the symbol after the SFD is selected as the DDMP</u>        | <u>90.5.2</u> | <u>When the MAC Merge sublayer is instantiated and the beginning of the symbol after SFD is selected as the DDMP, the value of the MM parameter of the TS_RX.indication primitive is set to EMAC if an SMD-E preceded the DDMP and to PMAC if an SMD-S preceded the DDMP.</u>                                               | <u>MM:M</u> | <u>Yes [ ]</u><br><u>N/A [ ]</u> |

*Change the title and table in 90.8.4 as follows:*

**90.8.4 DDMP selection~~Data delay reporting~~**

| Item              | Feature                                | Subclause       | Value/Comment                                                                                                                                                                                                                                                    | Status       | Support            |
|-------------------|----------------------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------------|
| <del>Report</del> | <del>Data delay reporting format</del> | <del>90.7</del> | <del>Data delay measurement is reported in the form of a quartet of values; the maximum transmit data delay, the minimum transmit data delay, the maximum receive data delay, and the minimum receive data delay, for each MMD instantiated in the device.</del> | <del>M</del> | <del>Yes [ ]</del> |
| <u>DDMP</u>       | <u>DDMP selection</u>                  | <u>90.7</u>     | <u>The DDMP is either the beginning of the SFD or the beginning of the first symbol after the SFD (see 45.2.3.69a and 45.2.5.31).</u>                                                                                                                            | <u>M</u>     | <u>Yes [ ]</u>     |

## Annex A

(informative)

## Bibliography

*Change bibliography entries [B39] and [B41] as follows:*

[B39] IEEE Std 802.1AS™, IEEE Standard for Local and Metropolitan Area Networks—Timing and Synchronization for Time-Sensitive Applications in Bridged Local Area Networks.

[B41] IEEE Std 1588™-2008, IEEE Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems.

*Insert new Annex 90A after Annex 86A as follows:*

## **Annex 90A**

(informative)

### **Timestamping accuracy considerations**

#### **90A.1 Sub-nanosecond timestamping introduction**

This annex provides information on supporting sub-nanosecond timestamping for time synchronization protocol (TimeSync) Client implementations compliant with Clause 90. This timestamping may be used for time synchronization protocols including IEEE Std 1588 [B41] and IEEE Std 802.1AS [B39].

#### **90A.2 Sub-nanosecond timestamping background**

Ethernet support for time synchronization protocols (Clause 90) was not originally specified to support sub-nanosecond timestamping. Thus, implementation flexibility permitted by this standard prior to the addition of certain registers (Data delay measurement point ability, Multilane ability, and PCS dynamic path data delay ability registers shown in Table 45–293 and the sub-ns resolution path data delay registers located throughout 45.2.1 to 45.2.6) could lead to timestamp accuracy impairments that might not satisfy sub-nanosecond timing requirements.

Timestamping accuracy can be impaired when two TimeSync Clients do not account for variation in Physical Layer device (PHY) path data delay in the same manner. Examples of PHY functions that cause variation in the PHY path data delay include alignment marker or codeword marker insertion/removal, idle insertion/removal, and Physical Coding Sublayer (PCS) lane distribution/merging.

Timestamping accuracy is impaired when two TimeSync Clients do not use the same data delay measurement point. As specified in 90.7 and 90.4.3.1.1, there are two options for the data delay measurement point (the beginning of the start of frame delimiter (SFD), and the beginning of the first symbol after the SFD). The use of the beginning of the first symbol after the SFD is consistent with IEEE Std 1588 [B41] and IEEE Std 802.1AS [B39].

Table 90A–1 shows the magnitude of potential timestamp accuracy impairments that could be generated by the aforementioned causes.

#### **90A.3 Considerations for use of different data delay measurement points**

If two TimeSync Clients use different data delay measurement points, a timestamp accuracy impairment equal to the time difference between the two data delay measurement points will be incurred on the TimeSync link delay measurement. The magnitude of this impairment, as shown in Table 90A–1, is the time difference between the beginning of the SFD data delay measurement point and the beginning of the first symbol after the SFD data delay measurement point when they are adjacent to each other, which they normally are. If the PCS dynamic path data delay ability and multilane ability are not in use (see 90.7, 90.5.3, 90.5.4, Table 45–293, and Table 45–295a), an additional impairment could result if these two data delay measurement point options are further separated due to:

- Insertion of octets between the two data delay measurement points for alignment marker or codeword marker functions
- Multiple PCS lane distribution delays

**Table 90A–1—Magnitude of potential timestamp accuracy impairments**

| Ethernet rate | Magnitude of potential timestamp accuracy impairments per transmit or receive port (ns) |                                          |                                                                    |                                         |
|---------------|-----------------------------------------------------------------------------------------|------------------------------------------|--------------------------------------------------------------------|-----------------------------------------|
|               | Mismatched data delay measurement point <sup>a</sup>                                    | Idle insertion / removal <sup>b, c</sup> | Alignment marker/ codeword marker insertion / removal <sup>c</sup> | PCS lane distribution / merging         |
| 10M           | 800                                                                                     | 400                                      | N/A                                                                | N/A                                     |
| 100M          | 80                                                                                      | 40                                       |                                                                    | N/A                                     |
| 1G            | 8                                                                                       | 16 <sup>d</sup> , 8 <sup>e</sup>         |                                                                    | 0 <sup>e</sup> , N/A <sup>d</sup>       |
| 2.5G          | 3.2                                                                                     | 12.8                                     |                                                                    | N/A <sup>g</sup>                        |
| 5G            | 1.6                                                                                     | 6.4                                      |                                                                    | N/A <sup>g</sup>                        |
| 10G           | 0.8                                                                                     | 3.2                                      |                                                                    | N/A <sup>d, f, g</sup> , 0 <sup>e</sup> |
| 25G           | 0.32                                                                                    | 1.28                                     | 10.24                                                              | N/A                                     |
| 40G           | 0.2                                                                                     | 1.6                                      | 6.4                                                                | 4.8                                     |
| 50G           | 0.16                                                                                    | 1.28                                     | 5.12                                                               | 3.84                                    |
| 100G          | 0.08                                                                                    | 0.64                                     | 12.8                                                               | 12.16                                   |
| 200G          | 0.04                                                                                    | 0.32                                     | 2.56                                                               | N/A <sup>g</sup>                        |
| 400G          | 0.02                                                                                    | 0.16                                     | 2.56                                                               | N/A <sup>g</sup>                        |

<sup>a</sup> The value shown only accounts for the time between the two data delay measurement point options when they are adjacent. See 90A.3 for other factors that affect some of these values.

<sup>b</sup> The value shown corresponds to the effect of a single idle insertion/removal.

<sup>c</sup> The path data delay of a packet can be affected when its data delay measurement point occurs after an alignment marker, codeword marker, or idle insertion/removal event.

<sup>d</sup> For 1000BASE-X or 10GBASE-R.

<sup>e</sup> For 1000BASE-T or 10GBASE-X.

<sup>f</sup> For 10GBASE-T.

<sup>g</sup> For PHYs including FEC, the lane distribution/merging operation belongs only to the forward error correction (FEC) function. The FEC lane distribution/merging operation is not subject to potential timestamp accuracy impairments because its path data delay determination is already clearly defined, and not subject to implementation flexibilities.

Implementations with sub-nanosecond resolution path data delay reporting abilities (see 45.2.1.175, 45.2.2.20, 45.2.3.67, 45.2.4.28, 45.2.5.28, and 45.2.6.14) and path data delay signaling (see 90.5.3 and 90.5.4) suffer a timestamp accuracy impairment of only one octet time between the two data delay measurement point options because:

- The effect of alignment marker or codeword marker insertion is accounted for, using the TX\_NUM\_BIT\_CHANGE and RX\_NUM\_BIT\_CHANGE signals (see 90.5.3, 90.5.4, and 90A.5).
- The multiple PCS lane path data delay is modeled as a constant value for all PCS lanes (see 90.7.1 and 90A.4).

## 90A.4 Considerations for multiple PCS lane functions

The general concept used to accommodate the delay variation of the multiple PCS lane distribution/merging operation is explained in 90A.7. This concept takes advantage of the fact that the sum of the delay variations that are intrinsic to the transmit multiple PCS lane distribution operation or to the receive multiple PCS lane merging operation is a predetermined constant for the given multiple PCS lane function (when the multilane ability (3.1800.11) bit is set, see 45.2.3.67.2).

This concept allows the intrinsic delay variations to be modeled as constant values, thus enabling the static TimeSync PCS transmit path data delay register and TimeSync PCS receive path data delay register to be used with sub-nanosecond timestamping even when multiple PCS lane functions are present. As explained in 90.7.1, the TimeSync PCS transmit path data delay register would use the greatest PCS lane distribution delay as its constant value (which corresponds to the start of the transmit PCS lane distribution function) and the TimeSync PCS receive path data delay register would use the smallest PCS lane merging delay as its constant value (which corresponds to the start of the receive PCS lane merging function).

Because the PCS transmit path data delay is modeled as a constant value, the minimum and maximum TimeSync PCS transmit path data delay registers in an ideal implementation have the same value due to the multiple PCS lane distribution operation. Likewise, because the PCS receive path data delay is modeled as a constant value, the minimum and maximum TimeSync PCS receive path data delay registers in an ideal implementation have the same value due to the multiple PCS lane merging operation. Having identical minimum and maximum values in these registers indicates that there is no uncertainty in the PCS path data delay.

## 90A.5 Considerations for alignment marker/codeword marker and idle functions

Timestamp accuracy impairment can occur because alignment marker, codeword marker, or idle insertion and removal events cause an instantaneous change in the PCS path data delay. Unlike other PHY functions, these events generate PCS path data delay variations that do not result in a constant sum of the transmit and receive delays, and thus cannot be predetermined.

Each of these path data delay variations may be accounted for by using the TX\_NUM\_BIT\_CHANGE and RX\_NUM\_BIT\_CHANGE signals (see 90.5.3 and 90.5.4). These signals allow the TimeSync Client to compensate for the instantaneous change in the path data delay. Because the signals compensate for the instantaneous path data delay changes, the TimeSync PCS transmit path data delay register and TimeSync PCS receive path data delay register can operate as static values, even when alignment marker, codeword marker, or idle insertion/removal operations are present.

Examples that show how TX\_NUM\_BIT\_CHANGE and RX\_NUM\_BIT\_CHANGE may be used are given in 90A.5.1 and 90A.5.2, respectively.

### 90A.5.1 Example use of TX\_NUM\_BIT\_CHANGE and PDPDD

- a) Scenario without alignment marker, codeword marker, or idle insertion/removal event:
  - i) Arrival time of a data delay measurement point at the transmit xMII =  $T_1$ .
  - ii) Transmit PCS path data delay = PDD1. The constant value, PDD1, is programmed into the TimeSync PCS transmit path data delay registers.
  - iii) Calculated transmit departure timestamp =  $T_1 + \text{PDD1}$ .
- b) Scenario with alignment marker, codeword marker, or idle insertion/removal in which the PDPDD parameter, which is conveyed by TX\_NUM\_BIT\_CHANGE, is used to account for the transmit

PCS path data delay variation, allowing the transmit PCS path data delay to be modeled as a constant:

- i) Adjusted arrival time of the data delay measurement point at the transmit  $xMII = T1 + PDPDD \times \text{bit time}$ .
- ii) The arrival time at the transmit  $xMII$  is modified to reflect the alignment marker, codeword marker, or idle insertion/removal event (as if it happened before the transmit  $xMII$ , per 90.7.2).
- iii) The value of PDPDD is positive when alignment marker, codeword maker, or idle is inserted ahead of the data delay measurement point, increasing the transmit path data delay, and negative when idle is removed ahead of the data delay measurement point, decreasing the transmit path data delay.
- iv) Transmit PCS path data delay = PDD1.
- v) The constant value, PDD1, programmed into the TimeSync PCS transmit path data delay registers does not change.
- vi) Calculated transmit departure timestamp =  $T1 + (PDD1 + PDPDD \times \text{bit time})$ .

### 90A.5.2 Example use of RX\_NUM\_BIT\_CHANGE and PDPDD

- a) Scenario without alignment marker, codeword marker, or idle insertion/removal event:
  - i) Arrival time of a data delay measurement point at the receive  $xMII = T2$ .
  - ii) Receive PCS path data delay = PDD2. The constant value, PDD2, is programmed into the TimeSync PCS receive path data delay registers.
  - iii) Calculated receive arrival timestamp =  $T2 - PDD2$ .
- b) Scenario with alignment marker, codeword marker, or idle insertion/removal in which the PDPDD parameter, which mirrors the corresponding value of RX\_NUM\_BIT\_CHANGE, is used to account for the receive PCS path data delay variation, allowing the receive PCS path data delay to be modeled as a constant:
  - i) Adjusted arrival time of the data delay measurement point at the receive  $xMII = T2 - PDPDD \times \text{bit time}$ .
  - ii) The arrival time at the receive  $xMII$  is modified to reflect the alignment marker, codeword marker, or idle insertion/removal event (as if it happened after the receive  $xMII$ , per 90.7.2).
  - iii) The value of PDPDD is positive when idle is inserted ahead of the data delay measurement point, increasing the receive path data delay, and negative when alignment marker, codeword maker, or idle is removed ahead of the data delay measurement point, decreasing the receive path data delay.
  - iv) Receive PCS path data delay = PDD2.
  - v) The constant value, PDD2, programmed into the TimeSync PCS receive path data delay registers does not change.
  - vi) Calculated receive arrival timestamp =  $T2 - (PDD2 + PDPDD \times \text{bit time})$ .

### 90A.5.3 Considerations for implementations without TX\_NUM\_BIT\_CHANGE and RX\_NUM\_BIT\_CHANGE

For an implementation that does not compensate for the path data delay variation resulting from alignment marker, codeword marker, or idle insertion/deletion removal events (e.g., without the TX\_NUM\_BIT\_CHANGE and RX\_NUM\_BIT\_CHANGE signals), the effect of the timestamp accuracy impairments that result from these events may be evaluated to determine if they cause significant degradation in the TimeSync system's performance. Some observations that might help this evaluation are given below:

- Typically, the probability that an alignment marker, codeword marker, or idle insertion/deletion removal event affects the path data delay of a packet is small.

- A low-pass filter, which might be present in a TimeSync Client's time recovery algorithm, could attenuate the effect of the resulting impairments.
- An implementation that does not transmit packets in the region of alignment marker/codeword marker insertions or idle insertions/removals might avoid the generation of the impairment at its transmit port. However, this does not guarantee that the corresponding remote receive port will not generate impairments from its own idle insertions/removals.

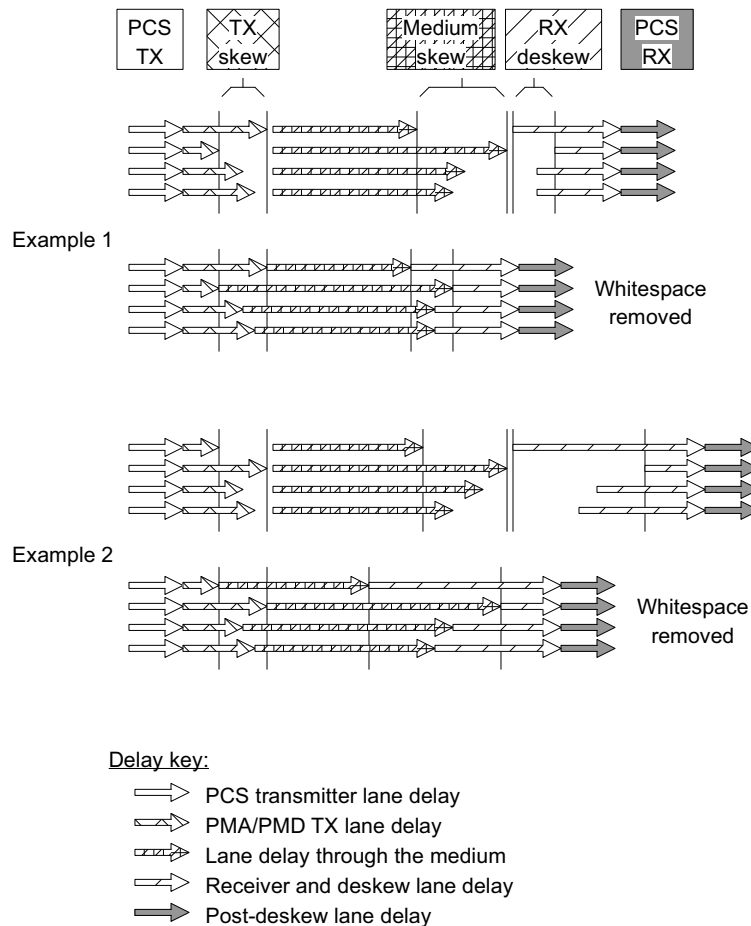
If a PHY does not perform insertion or removal of alignment markers, codeword markers, or idles, then the TX/RX\_NUM\_BIT\_CHANGE signals are not needed.

## 90A.6 Considerations for transmit skew

For a multi-lane PHY, the receiver accounts for the skew of the medium by timestamping with respect to the lane with the smallest deskew buffer delay (see 90.7.3). Thus, the medium delay used by the time synchronization protocol is that of the lane with the greatest delay.

For these multi-lane PHYs, it is difficult to compensate for the presence of skew at the transmit Medium Dependent Interface (MDI) because this skew is entwined with, but independent from, the skew of the medium. As shown in the examples of Figure 90A–1, the transmit skew in series with the medium skew is either additive or subtractive. Example 1 and Example 2 in Figure 90A–1 have the same transmit skew and the same medium skew but, because these skews are associated differently, the total skew seen at the receiver is different. By obscuring the latency of the medium, transmit skew can contribute to time synchronization error.

The per-lane transmit skew values might be compensated at the receiver, where the total skew of each lane can be observed at its receive deskew buffers. By using the observed per-lane total skew values at the receiver and the per-lane transmit skew values, the actual skew of each lane of medium could be determined. To avoid the need for this type of processing, it is recommended that multi-lane transmitter implementations try to minimize the lane skew at their MDI.



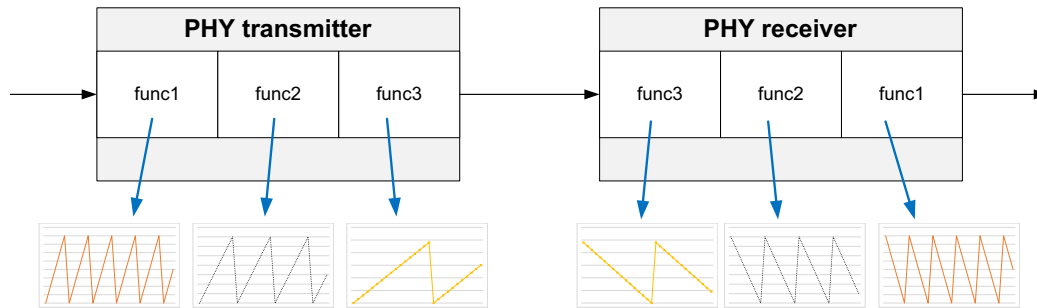
**Figure 90A-1—Transmit PMA/PMD skew in series with medium skew**

## 90A.7 General method for dealing with repeating delay variation patterns

Many PHY functions have intrinsic delay variations with the following characteristics:

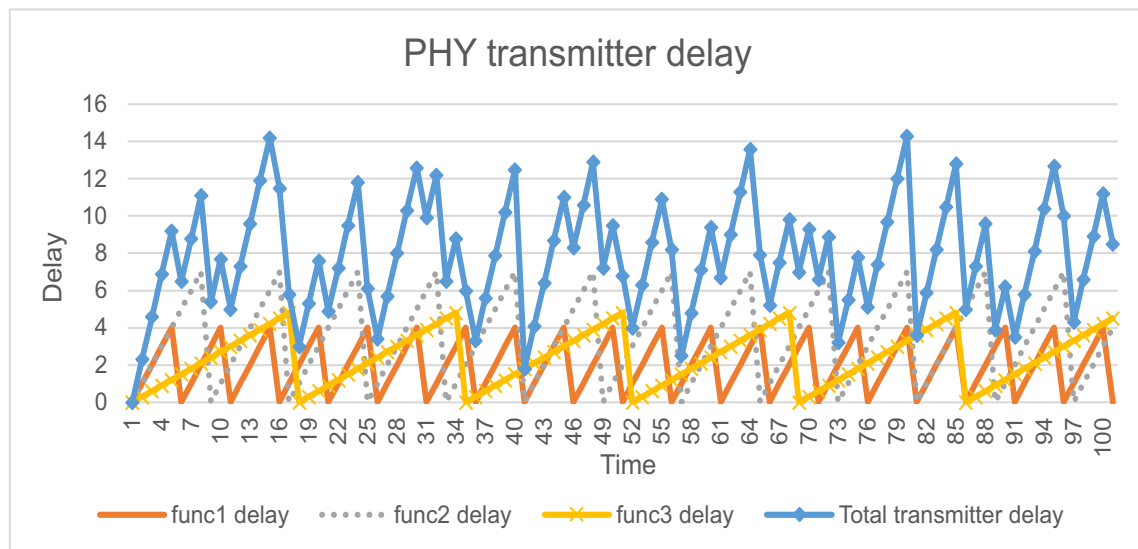
- The transmit and the receive intrinsic delay variations follow a known repeating pattern.
- The transmit intrinsic delay variation pattern is a mirror of the receive intrinsic delay variation pattern and the sum of the two intrinsic delays is a known constant value. This is true because the data stream before the transmit function and the data stream after the receive function are identical.

It is possible to take advantage of the above characteristics to simplify the path data delay modeling. For example, if a PHY has multiple functions with these delay characteristics, as shown in Figure 90A-2, its aggregated path data delay can be modeled as a constant value instead of the dynamically varying sum of multiple varying delays.



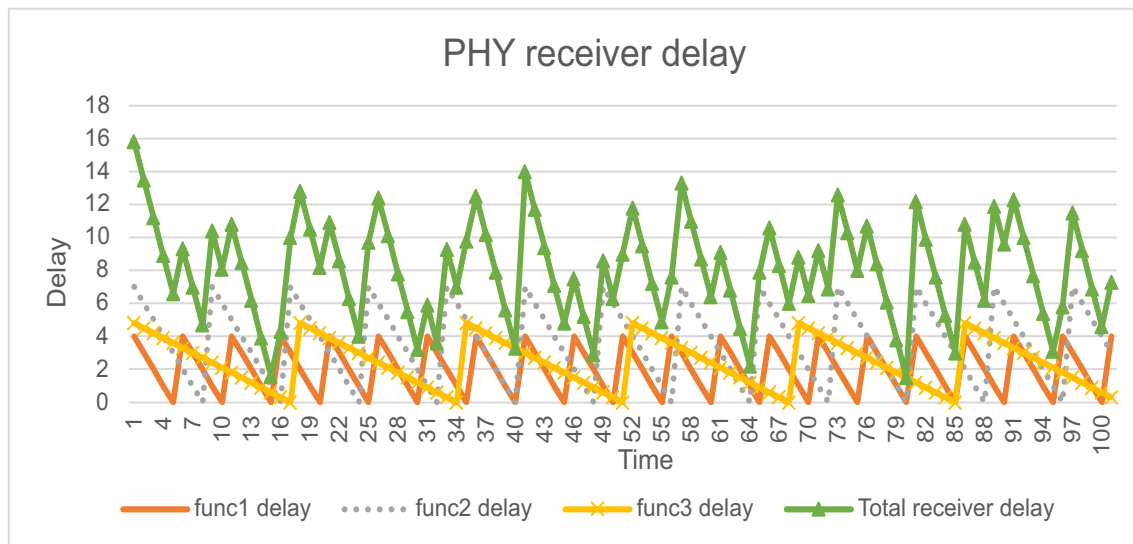
**Figure 90A-2—PHY with cascaded functions with varying delays**

For the example shown in Figure 90A-2, the delay of each transmit function and the sum of them are shown in Figure 90A-3 and the delay of each receive function and the sum of them are shown in Figure 90A-4. These sums have no easily discernible pattern and might require an implementation to determine the instant path data delay for any chosen bit that corresponds to the data delay measurement point of a packet in the Ethernet data stream.



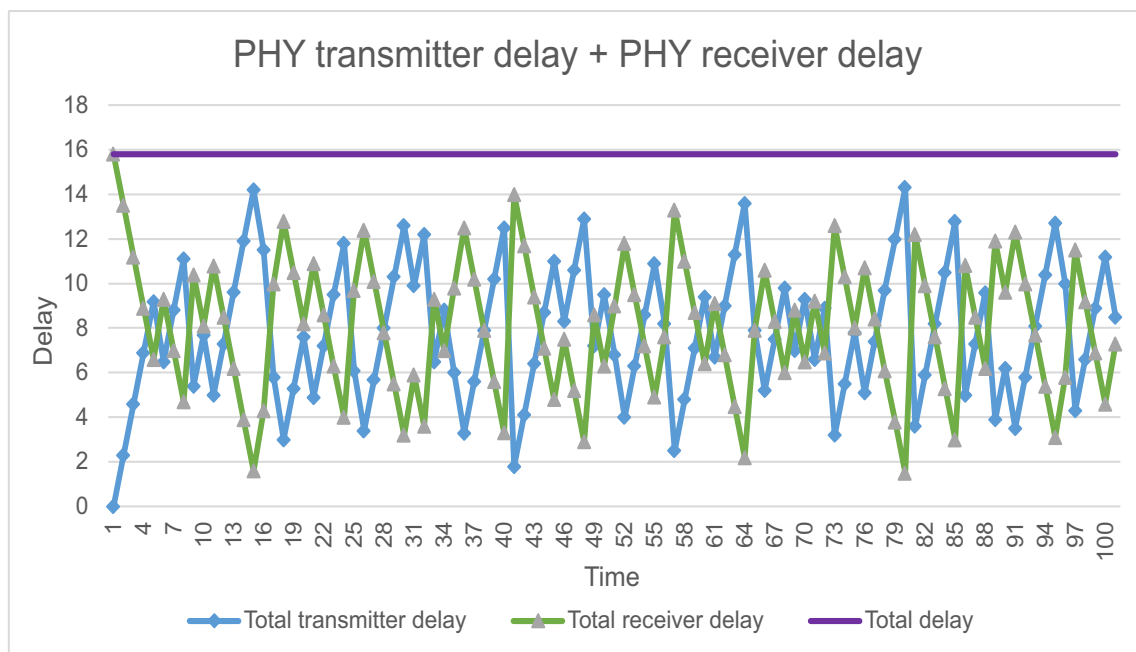
**Figure 90A-3—Total delay of transmit PHY with cascaded varying delays**

Because the intrinsic delay variation in the receive PHY is a mirror of the intrinsic delay variation in the transmit PHY, the total intrinsic delay through both PHYs is a constant, as illustrated in Figure 90A-5. This eliminates the need to track the varying delay of the data delay measurement point of a packet through the transmit PHY and the receive PHY. Instead, it is possible to divide the aggregate constant delay into transmit and receive portions and allocate them to the individual PHY instances.



**Figure 90A-4—Total delay of receive PHY with cascaded varying delays**

The allocated portions of the constant total intrinsic delay value are then added to the implementation-specific delays, which are also constant values, of the corresponding transmit and receive PHYs to compensate these intrinsic delay variations into timestamps.



**Figure 90A-5—Transmit PHY delay, receive PHY delay, and total delay**

It is recommended to use this method to model all PHY delay variations of this nature. When using this method, it is also recommended to remain consistent with the FEC (see 90.7.1) and the multiple PCS lane distribution/merging functions (see 90.7.1 and 90A.4) and allocate the maximum value of the intrinsic delay to the transmit PHY and the minimum value of the intrinsic delay to the receive PHY. A couple of common PHY functions that have delay variations of this nature but that are not dealt with specifically in this

standard are 64B/66B encoding/decoding and pulse amplitude modulation (PAM) encoding/decoding. There might be others.

The example shown in Figure 90A–3, Figure 90A–4, and Figure 90A–5 have the maximum delays of all transmit functions and the minimum delays of all receive functions aligned at the beginning of time. This ideal alignment will occur periodically if at least one of the following scenarios is true:

- a) The alignment of the transmit (or receive) functions is enforced by a specified relationship between them.
- b) The transmit (or receive) functions' periods are unrelated, that is, the maxima (and minima) of the transmit (or receive) functions' latency are guaranteed to coincide eventually.

When there is no fixed alignment between transmit (or receive) functions, and the functions' periods are related by a simplified ratio of  $N/M$ , where  $N \leq M$ , the maximum (or minimum) aggregated delay of the functions is not the simple sum of their individual maximum (or minimum) delays. Instead, the maximum (or minimum) aggregated delay depends on the in-situ alignment of the functions. How this aggregated delay is determined and accounted for in the transmit (or receive) path data delay value is beyond the scope of this standard. However, note that the maximum potential difference between the actual aggregated delay and the summation of the individual delays of the functions can be calculated. In this scenario, the maxima (or minima) of the first function's variable latency coincides to the maxima (or minima) of the second function's variable latency to within  $1/(2N)$  of the shorter function's period. Whether the potential delay difference between the aggregated delay and the sum of the individual function delays is small enough to satisfy the timing requirements is up to the individual application.

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